Straw Burns In Fields, Forever?: Displacement Effects of a Seasonal Burning Ban in Thailand*

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Abstract

Outdoor air pollution from biomass burning poses a significant health and economic challenge in Southeast Asia. Yet, authorities in low-income settings face limited capacity to enforce environmental regulations. Targeted restrictions that focus on high-risk seasons or locations can help address that, by concentrating resources on specific periods or places. This paper examines the effect of seasonal burning bans implemented in Northern Thailand from 2013. Using a synthetic control method, I find that in periods of relatively strict enforcement, the policy reduced satellite-detected fires by 28%. However, the bans caused temporal displacement, with fire activity increasing after they ended. Additionally, they led to shifting of burning times to avoid detection, which can explain at least a tenth of the estimated reduction in fires. These findings suggest that while targeted regulations can reduce harmful activities, their effectiveness may be undermined by behavioural responses and lead to displacement.

JEL Codes: K32, K42, Q53, Q54, Q58

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1 Introduction

Outdoor air pollution is one of the leading causes of deaths globally. It has caused an estimated 4.2 million premature deaths worldwide in 2018, nearly a third of those in Southeast Asia (World Health Organization, 2018). Pollution levels in much of Southeast Asia peak annually during the dry season, from around January to the end of April. The extreme levels of pollution in parts of Southeast Asia during this period are driven by an increase in some polluting activities, combined with metrological and atmospheric conditions that trap pollution close to the ground and prevent its dispersion. The main polluting activities that increase during the dry season are agricultural and forest fires (Fotiou & Perkins, 2021; Moran et al., 2019). Agricultural fires are often set up by farmers several weeks after harvesting to cheaply dispose of crop residue, before the next seeding. Forest fires, meanwhile, are often set up in Southeast Asia to support collection of forest products such as valuable mushrooms at the end of the dry season (Pirard & Charoenpanwutikul, 2023).

In Northern Thailand, one of the regions with the highest levels of pollution, particulate pollution is approximately six times higher than the WHO guideline. This leads to adverse health consequences. Residents of Chiang Mai province - home to Thailand's second most populated city - are estimated to gain 2.4 years in life expectancy on average, if pollution levels were reduced to those of the WHO's guideline (Air Quality Life Index, 2019). The extreme pollution also hurts tourism, a major source of revenue in many regions in Southeast Asia, and leads to school closures, disruption of aerial transportation, lower productivity, and other adverse outcomes (Evrard & Mostafanezhad, 2023; Voiland, 2024).

Yet, the ability of governments in low-income settings to effectively regulate environmental externalities is often limited. Targeted enforcement, where resources are focused on locations or periods with higher risk or damages, can alleviate these constraints by reducing and concentrating enforcement costs. Seasonal restrictions, for example, are used to manage fishing in China (Yuan, 2025), and hunting in Ghana (Adom & Boamah, 2020). A critical challenge is that imperfect enforcement may displace illegal activities to times or places with weaker enforcement or no restrictions. While seasonal regulations are a promising tool, their impact remains not well understood.

This paper evaluates a seasonal moratorium on open burning, established in Northern Thailand starting from 2013. In response to the high pollution in the haze season, nine provinces in Northern Thailand have set seasonal burning bans, with the bans usually lasting two months every year from around February to April. The provinces were responsible for setting the dates for each annual ban and for enforcing it. Initially, enforcement was limited, but it became stricter starting in 2017 (Mostafanezhad & Evrard, 2021; Pirard & Charoen-

panwutikul, 2023). Monitoring of the ban was done in part using FIRMS, a NASA-run fire information system that detects fire hotspots using thermal anomalies from remote sensing (Smith, 2024).For my analysis, I utilize observations from the MODIS sensor, which is mounted on the Aqua and Terra satellites and overpasses Thailand every day at 1:30 and 10:30 (AM and PM).

I will evaluate the effect of the ban using a synthetic control method (Abadie, 2021). This method constructs a weighted average of donor (never treated) units that closely matches the pre-intervention outcomes of each treated unit. Then, differences in post intervention outcomes can be attributed to the treatment. The treated units are the nine provinces in Northern Thailand with seasonal burning bans, whereas the donor units are composed of districts in Myanmar, provinces in Laos, and provinces in the rest of Thailand. I construct the weights for the synthetic controls using the number of fires, land use, elevation, and climatic variables from before 2013, the start of the "soft" moratorium. Given that there are multiple treated and donor units, I use the penalised synthetic control proposed by Abadie and L'Hour (2021) to guarantee unique, optimal weighted combination of donor units for each treated unit. The effects of the ban are evaluated for the months of January to May, as very few fires take place outside of this period. I evaluate the effects separately for the period of weak enforcement of 2013-16, and the period with stricter enforcement of 2017-20. To evaluate heterogeneity and the possibility of displacement of fires throughout the year, I construct month- or outcome-specific synthetic controls.

I do not find a significant effect of the ban between 2013-2016, during the period when enforcement was limited. Meanwhile, from 2017 to 2020, when the ban was more strictly enforced, the number of detected fires per 100 km ^{2} from January to May decreased by 3.24. This is equivalent to 3,419 fewer fires per year in all of Northern Thailand, or a reduction of 28% compared to the synthetic versions. The decrease is primarily a result of a reduction in fires around March, when burning is typically prohibited in all of Northern Thailand's provinces. However, the burning ban contributes to an extended burning season. When examining the monthly effects, I find that the number of fires detected in May - after the end of the dry season - increases by 0.29 per 100 km^2 (305 fires across Northern Thailand). This corresponds to 373% more fires in May compared to the synthetic controls. To also account for bans ending during April, I utilise the known dates of the seasonal bans and evaluate effects in half-months (two weeks) depending on the treatment status. I find that in the half-month after the end of the ban, the number of fires in Northern Thailand rises by 1.77 fires per 100 km^2, approximately 1,862 fires across the nine treated provinces. The reduction is primarily driven by fires taking place during the day, particularly those detected by the Aqua satellite's daytime overpass. The reduction in fires is also mainly associated with high-intensity fires, and with fires taking place in areas that are forested, have low

agricultural output, and located far from urban settlements and roads.

While the reduction may reflect compliance with the seasonal bans and genuine reduction in fires, it can also reflect attempted evasion from satellites. As the Thai authorities use FIRMS to monitor fires during the ban, villagers who wish to burn can adjust the burning times to minimize detection risk. This can be done by burning in the nine-hour window between the overpass of the Aqua satellite and the Terra satellite, if the fire extinguishes by Terra's overpass. I observe that the share of fires detected in Aqua's daytime overpass decrease during the ban, consistent with temporal shifting to avoid detection by satellites. I assume that the share of fires detected in Aqua's daytime overpass should be the same before the ban and out of the total documented reduction during the ban. Using that, I estimate that during the strict enforcement period of 2017-2020, at least 10.6% of the estimated reduction in detected fires is likely attributed to evasion from satellites. This suggests that while remote sensing can reduce the costs of monitoring illegal activities, its effects could be hampered by behavioural responses. This is particularly relevant in cases, like crop and forest burning, where the illegal activity can be short-lived and not be captured by common remote sensing tools due to their temporal and spatial resolutions.

This paper contributes to the growing literature on policies aimed at mitigating forest and agricultural fires, particularly in low- and middle-income countries. Recent studies examine how economic conditions and incentives faced by individuals shape burning behaviour (Behrer, 2023; Garg et al., 2024; Jack et al., 2024). Various works examine public policies intended to discourage burning, such as providing farmers with conditional cash payments for not burning (Edwards et al., 2020; Jack et al., 2024); expanding access to biomass power plants, which reduce disposal costs for crop residues (Cao & Ma, 2023); investigating firms involved in burning (Balboni et al., Forthcoming); restricting movements in forests (Paudel, 2021); awareness campaigns (Edwards et al., 2020; Nguyen Huy et al., 2024); and supporting fire prevention infrastructure (Nguyen Huy et al., 2024). Few studies assess the effects of prohibitions on burning, often finding limited effects due to weak enforcement capacity (Cao & Ma, 2023), misaligned bureaucratic incentives (Dipoppa & Gulzar, 2024), and potentially corruption (Balboni et al., Forthcoming). Closest to my work, Cao and Ma (2023) show that burning bans in China only reduce nighttime fires, when patrol-based detection is easier, suggesting that the effectiveness of bans depends on the cost of enforcement. My work add to those by providing evidence on the impact of a seasonal fire moratorium, a policy that may be easier to enforce or implement for political economy reasons. I show that such a policy, while reducing fires in the peak of the burning season, can also lead to a temporal displacement of fire activity.

Second, this work joins the literature on environmental regulation in lower income settings

(e.g., Ashraf et al. (2016) and Greenstone and Hanna (2014)). Digital tools and remote sensing can improve the limited capacity of governments in those regions to quickly and accurately detect illegal activities (Saavedra, 2025). Recent papers evaluate the effectiveness of using such tools in various contexts, including tax audits (Bachas et al., 2025), illegal mining (Saavedra, 2025), and deforestation (Assunção et al., 2023; Moffette et al., 2021). My work adds to this literature by studying a fire ban in Thailand that was supported by an alert system based on remote sensing. I show that the impact of such systems can be undermined by technological limitations and behavioural responses to monitoring. As individuals can learn about the patterns of monitoring system, such as the predictable overpass times of satellites, they may adjust their behaviour to decrease the risk of being detected. This highlights a key limitation of enforcement using remote sensing, particularly when the illegal activities can be brief, and when accountability is constrained by ambiguous property rights are.

The remainder of the paper is structured as follows. Section 2 provides background on the haze season and Thailand's seasonal burning bans. Section 3 details the data used, while Section 4 outlines the empirical strategy. The results and robustness tests are presented and discussed in Section 5, while Section 6 presents an attempt to quantify the share of reduction in fires that could be attributed to detection avoidance. Section 7 concludes.

2 Context

2.1 Haze season in Southeast Asia

Biomass burning is a significant contributor to the high levels of pollution in Southeast Asia from around January-February to the end of April (Fotiou & Perkins, 2021). Figure 1 shows the number of average fires per province in Laos, Myanmar, Northern Thailand, and the rest of Thailand. The dry season consistently features extreme levels of burning compared to the rest of the year, commonly known in Northern Thailand as the "haze season". The high pollution levels are driven by an increase in polluting activities, mainly the burning of crops, grasslands, and forests. These are exacerbated by specific meteorological conditions during this period of the year, such as limited precipitation and low winds, which trap pollution close due to the ground due to thermal inversion (Moran et al., 2019; Mostafanezhad & Evrard, 2021; Pirard & Charoenpanwutikul, 2023).

One of the main sources of fires in the region is agricultural fires. After harvesting, farmers need to dispose of crop residue (waste materials) before seeding the next crops. Although there are mechanical disposal alternatives, burning remains the cheapest way to dispose of crop residue around much of Southeast Asia, in particular in the face of liquidity constraints.

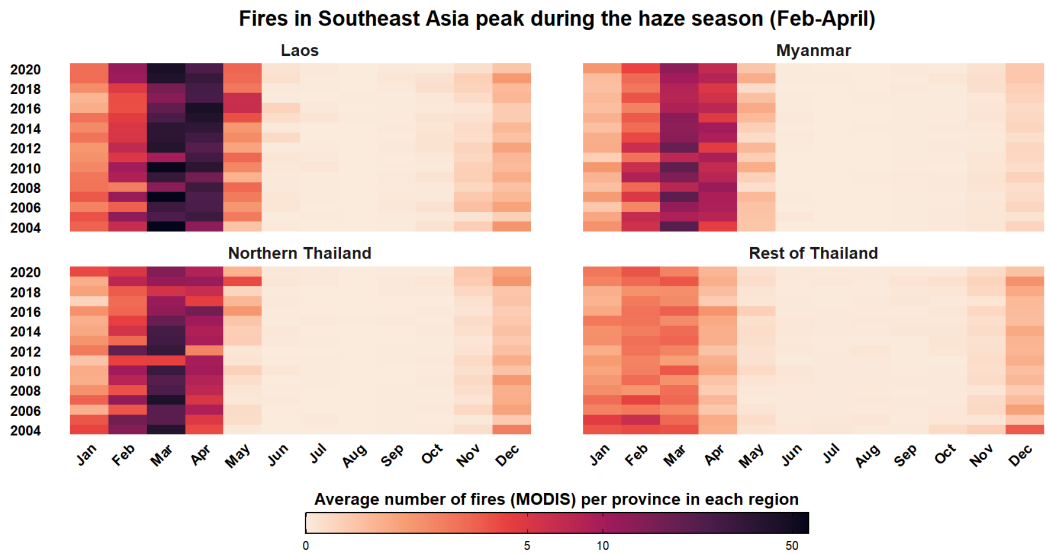


Figure 1: Fires over time by region

The choice to burn crop residue is largely driven by its low cost, as well as farmers' limited access to machinery or labour that would allow them to opt for alternative disposal methods (Adeleke et al., 2017). The choice to burn is also driven by the belief of some farmers that it generates positive effects for the soil, such as promoting tilling and controlling insects, diseases, weeds, and animals in the fields. However, in addition to its environmental and health impacts, burning can also negatively effect soil fertility and require greater use of fertilizers (Pirard & Charoenpanwutikul, 2023). Farmers are generally aware that burning causes pollution and adverse health consequences. Thus, several rural communities in Northern Thailand have self-governing rules that restrict open-burning. However, compliance with these rules is imperfect, especially as communities suffer from pollution from neighbouring countries and regions due to burning there, implying that self-enforcement has a limited effect on reducing pollution (Adeleke et al., 2017).

Forest burning is also common in Southeast Asia and greatly contributes to the pollution during the haze season. The most common types of trees in Northern Thailand (dipterocarp and mixed deciduous) shed their leaves during the dry season. In forests composed of these trees, burning is mainly done to collect forest products, notably "earthstar" mushrooms. These mushrooms (which are sold for about 9-15 USD / kg) grow in forests and are collected in May and June, with many locals believing that burning forests during the dry season accelerates the growth of those mushrooms. In addition, fires clear leaves and grass in forests, easing the collection of mushrooms and other forest products (Pirard & Charoenpanwutikul, 2023). Some of the forest burning is caused by clearing land for agriculture, estimated to be less than 5-15% of the causes of forest burning in Northern Thailand (Outapa & Ivanovitch,

2019; Pirard & Charoenpanwutikul, 2023)

2.2 Northern Thailand burning ban

The Thai government has recognized the environmental and health damages of the haze season, and attempted to mitigate them. Starting in January 2013, nine provinces in the region of Northern Thailand were instructed to implement moratoriums on open burning for 60-90 days every year. The provinces are Chiang Mai, Chiang Rai, Lampang, Lamphun, Mae Hong Son, Nan, Phayao, Phrae, and Tak (Lualon et al., 2013; Nikam et al., 2021). Provincial authorities were responsible for setting the specific dates, which usually fell between mid-February and mid-April, or from the beginning of March to the end of April. Initially, enforcement was limited. A detailed table of the dates of most bans from 2015 to 2020 is shown in Appendix B. However, until the end of 2016, the enforcement of the ban was limited. In light of the limited effect of the bans, a stricter enforcement regime was established starting in 2017. Before the ban period in each year, the authorities carried out preventive actions such as raising awareness of the damages of burning and the penalties associated with violating the ban (Pirard & Charoenpanwutikul, 2023). During the ban, local authorities monitored the fires and attempted to enforce it (Nikam et al., 2021). The penalties for violations stood at up to 100k Baht (around 3000 USD), with a potential imprisonment of up to seven years. This has cause backlash from villagers (Mostafanezhad & Evrard, 2021). Following the end of the ban annually, some provinces granted certain farmers permission to start agricultural fires, if some sustainable actions are taken (Pirard & Charoenpanwutikul, 2023).

Enforcement of the ban remained imperfect even after 2017. This was due to limited enforcement capacity, villagers' backlash, and enforcement challenges stemming from a lack of well-defined property rights, especially in forest areas. In particular, some of the fires were set in forest areas where the Thai state does not recognize the property rights of rural residents, meaning that accountability for setting fires is limited (Mostafanezhad & Evrard, 2021). Monitoring of fire hotspots was conducted using various sources, including satellite imagery from NASA's FIRMS, which provided live alerts on potential fire spots to local authorities (Smith, 2024). However, as the satellites circulate over Northern Thailand at roughly the same hours each day, some farmers allegedly shifted their burning times to the early afternoon, between satellite overpasses, to avoid detection (Mostafanezhad & Evrard, 2021).

3 Data

3.1 Sources

Admin units - I aggregate data to the provincial level (administrative level 1 regions) in Thailand and Laos. For Myanmar, where provinces are generally considerably larger, I use districts (administrative level 2 regions) instead, to allow for better comparability. The boundaries are set using the R package *rnaturalearth* (South, 2017), while for Myanmar I use level 2 regions from Myanmar Information Management Unit (MIMU) (n.d.) This results in the sample including 68 provinces in Thailand, 80 districts in Myanmar, and 17 provinces in Laos. The average size of an admin unit is 8,100 km^2. The nine provinces in Northern Thailand that implemented seasonal fire bans from 2013 serve as the treated units, while Laos's provinces, Myanmar's districts and all other provinces in Thailand are the control units. The administrative units are shown in Figure 6, with the provinces of Northern Thailand shown in more detail in Figure 7.

Fires - My main outcome variable will be based of the number of fire spots in an administrative unit. I use data on fire spots from NASA's Fire Information for Resource Management System (FIRMS), which processes thermal anomalies detected by two remote sensing sensors, MODIS and VIIRS (NASA FIRMS, 2023a). I will focus on data obtained from the MODIS instruments mounted on the Aqua and Terra satellites, which detects thermal anomalies at a spatial resolution of 1 km^2. MODIS identifies potential fires when based on thermal signature anomalies when one of the two satellites passes over a given location, with each typically passing over Southeast Asia twice a day, Terra around 10:30 am and pm, and Aqua around 1:30 am and pm. The anomalies are filtered and classified into fire spots. The MODIS data includes information such as the fire's location, the detecting satellite, the timing of the observation, the confidence level that an observation represents fire, and the radiative power of the fire (which can be used to estimate the severity of a fire). Each observation represents one or more fire spots within a 1 km^2 pixel (NASA FIRMS, 2021). The VIIRS instrument offers a higher resolution (375m) and better night fire detection. However, its data is only available from 2012, from the S-NPP satellite that passes over Thailand around 1:30 AM and PM, which is too short to be used as the pre-treatment period. Therefore, my main specification uses MODIS observations for Thailand, Myanmar, and Laos from 2004 to 2020.

FIRMS has several limitations that could cause it to miss fire spots, especially ones taking place in small agricultural plots. First, its temporal resolution means that it can only capture fires that occur during satellite overpasses. This is especially problematic for detecting agricultural fires, which may last for only 2-3 hours in the region and therefore be missed (Pirard & Charoenpanwutikul, 2023)). Second, FIRMS may not detect fires due to

atmospheric interference such as cloud cover. This issue is not particularly worrying for my context, as cloud cover is typically limited during the dry season in Southeast Asia. Finally, FIRMS could fail to detect small or low intensity fires (NASA FIRMS, 2021; Walker, 2024). The average agricultural plot per household in Northern Thailand is around 8 hectares (0.08 km^2), often divided into plots of around a hectare each (Pirard & Charoenpanwutikul, 2023), which can fall below the detection threshold of MODIS.

I do not use the MODIS Fire CCI Burned Area product (Lizundia-Loiola et al., 2020; Padilla Parellada, 2018). This dataset includes information about burned area in a spatial solution of 250m, so it could capture some fires that are not detected by FIRMS due to strategic shifting of the burning. However, it is mainly useful to detect fires that spread out of control, but not fires that were started and contained (Boschetti et al., 2019; Nguyen Huy et al., 2024). As many fires in this context, and especially agricultural fires in small plots, are contained, I do not use this product.

Climate - I obtain monthly climatic data from the ERA5 dataset, produced by the European Centre for Medium-Range Weather Forecasts (Copernicus Climate Change Service, 2017). ERA5 provides aggregated monthly values for a range of climatic variables at a spatial resolution of approximately 11.1 km^2 at the equator. Using Google Earth Engine (Gorelick et al., 2017), I extract six variables: mean air temperature at 2 meters, mean dew point temperature (a proxy for the humidity of the air), total precipitation, mean surface temperature, and mean values of the u- and v-components of wind.[1] For each administrative unit, I calculated monthly averages of these variables for the period 2004–2012 by averaging in all ERA5 grid cells that fall partly or entirely within the boundaries of a unit.[2]

Elevation - I extracted elevation data from NASA JPL (2020). In some areas of Southeast Asia, such as Northern Thailand and eastern Myanmar, higher elevation areas tend to be inhabited by indigenous minority groups. These communities were often blamed for agricultural burning practices in Thailand (Mostafanezhad & Evrard, 2021). Due to their relative geographic remoteness, these communities may be less affected by fire ban policies.

Distances to roads and urban settlements - As I do not have data on the actual enforcement of the ban, I use proxies to measure enforcement capacity. I use measures of spatial accessibility from roads and urban settlements. Enforcement of fire bans requires a physical presence, meaning that locations more accessible from urban centres and major transportation networks are likely to experience stricter monitoring and faster response times. I define

[1] The u-component represents wind speed along west-east axis, while the v-component represents wind speed along north-south axis.

[2] I only use climatic data for the pre-2013 period, as the burning bans in Northern Thailand began in 2013. The climatic variables are pre-treatment characteristics used to obtain the optimal weights in the synthetic control, so they are not used after 2012.

urban areas using the 2010 Global Human Settlement - Settlement Model (GHS-SMOD) (Schiavina et al., 2023). I apply a binary distinction, grouping together areas classified as urban or suburban/peri-urban. Road network data was extracted from OpenStreetMap (2025), and filtered to only keep paved roads (OpenStreetMap contributors, 2025).

Crop production - I used the 2010 Spatial Production Allocation Model (SPAM) dataset (International Food Policy Research Institute, 2019) to differentiate crop production. For each 10 km^2 grid cell in the SPAM dataset, I assign a quantile ranking based on total agricultural production, to determine which units are more dependent on agriculture. To account for differences in seasonality of burning due to the use of different crops, I identify the main crop type in each grid cell. I then matched these characteristics to each fire in the FIRMS dataset based on its location.

Land use - To differentiate between agricultural fires and forest fires, I extract data about the land use in the proximity of fires from Potapov et al. (2022). I grouped land use classifications that refer to crops and forests. For each fire spot, I have computed the share of pixels belonging to crops and to forests in its bounding box depending on the sensor's spatial resolution (1km for MODIS and 375m for VIIRS). I define agricultural fires and forest fires as fires taking place in areas with at least 50% share of crop pixels, and tree pixels, respectively, within 1 km^2.

3.2 Descriptive statistics

Table 1 presents summary statistics regarding administrative units, split by region. The mean yearly number of fires per 100 km^2 in the nine provinces of Northern Thailand is 10.1. The mean area of the nine provinces is 117 units of 100 km^2, such that the total area of the region is 1,052 units of 100 km^2. Hence, this represents 10,625 fires per year on average between 2004-2020 in all of Northern Thailand, with the standard deviation equivalent to 3,366 fires per year. Most of the fires detected by MODIS in Northern Thailand are detected during the day (90%), and occur in areas I classify as forest areas (86%). The average district in Myanmar is relatively similar to the mean province in Northern Thailand, with slightly fewer fires, smaller area, and a higher dependence on agriculture. The mean province of Laos, meanwhile, has higher incidence of fires, mainly in forests. Compared to provinces in Northern Thailand, provinces in the rest of Thailand experience fewer fires, are smaller, more agricultural, and experience higher temperatures. Figure 2 shows the average number of fires per 100 km^2, normalized by the area of each unit.

Table 1: Summary statistics

Category	Country / Region			
	Laos	Myanmar	North Thailand	Rest of Thailand
*Yearly Fire Activity (MODIS)**				
Total Fires	16.5 (6.2)	8.93 (6.1)	10.1 (3.2)	3.90 (2.8)
Day Fires	15.7 (6.0)	8.21 (5.6)	9.15 (2.9)	3.45 (2.5)
Night Fires	0.81 (0.52)	0.72 (0.72)	0.95 (0.41)	0.45 (0.37)
Crop Fires**	0.20 (0.30)	0.64 (0.96)	0.47 (0.47)	2.29 (2.4)
Forest Fires**	15.6 (6.6)	7.18 (5.9)	8.73 (3.5)	0.86 (1.3)
Regional Characteristics				
Area (100 km^2)	135.0 (12.2)	83.8 (6.4)	117.0 (19.3)	60.4 (5.2)
Crop Share	0.05 (0.016)	0.22 (0.024)	0.10 (0.024)	0.34 (0.031)
Forest Share	0.87 (0.031)	0.58 (0.036)	0.76 (0.037)	0.34 (0.032)
Urban Share	0.01 (0.006)	0.04 (0.014)	0.04 (0.007)	0.10 (0.021)
Elevation (m)	660 (66)	471 (48)	578 (45)	149 (13)
Number of Provinces	17	80	9	68
Meteorological Conditions				
Temperature (C)	22.3 (1.7)	23.6 (2.7)	23.7 (0.9)	26.2 (0.9)
Precipitation (mm)	1,051 (133)	1,164 (370)	919 (77)	904 (218)

All values represent means of administrative units (provinces in Thailand and Laos, and districts in Myanmar) in each region. Standard deviations are in brackets.
*Fires represent number of fires detected by MODIS per 100 km^2.
**Crop fires represent fires taking place in areas where within 1 km ^{2}, more than 50% of the pixels are classified as agricultural by Potapov et al. (2022). Forest fires are equivalent, with pixels classified as trees.

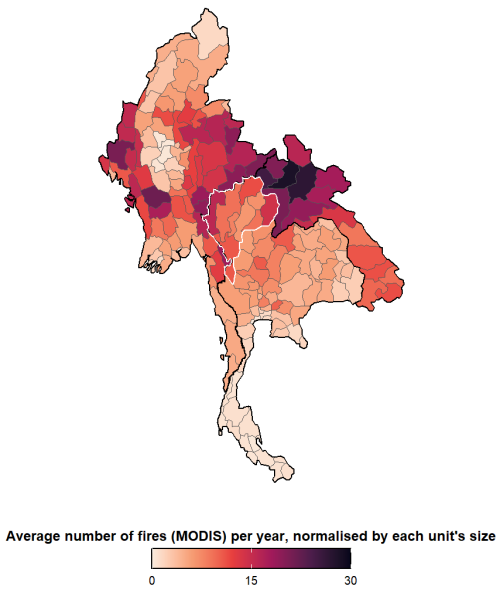


Figure 2: Map of the geographical units of analysis

Note: The figure shows the average number of fires per 100 km^2 in a year, normalised by each unit's area, in Thailand, Laos, and Myanmar. Fires are measured as total number of fires detected by MODIS in the years 2004-2020. The outer borders of the nine treated provinces in Northern Thailand are marked in white.

4 Empirical strategy

The ban on open burning was established in the Thai provinces with the highest fire incidence, making the selection into the treatment non-random. This regional differences due to climate, income and other characteristics, make the parallel-trends required for a difference-in-differences unlikely to hold. Therefore, I use a synthetic control method (SCM) (Abadie, 2021; Abadie et al., 2010) to evaluate the causal impact of the seasonal ban on open burning. SCM constructs a counterfactual for each treated unit by creating a weighted combination of untreated donor units. This method was used in many research topics, such as tobacco regulation (Abadie et al., 2010), immigration (Peri & Yasenov, 2019), taxation (Kleven et al., 2013), and, closest to my work, forest fire mitigation (Nguyen Huy et al., 2024). To account for the presence of multiple treated units, I use the penalized SCM proposed by Abadie and L'Hour (2021). This method adjusts the "classic" SCM for the case of multiple treated units, by adding a term λ that penalizes weights of donor units that are individually different to each treated unit. A more detailed explanation about the synthetic control method and the way I use it is available in Appendix A.

My goal is to estimate the causal effect of the ban, $\tau_{i,t} = Y^1_{i,t} - Y^0_{i,t}$, for each treated province i at time t. Here, $Y^1_{i,t}$ is the observed outcome under the ban, and $Y^0_{i,t}$ is the unobserved counterfactual outcome in the absence of the ban. The synthetic control method estimates

this counterfactual for i by constructing a weighted combination of untreated donor units that closely resemble the pre-intervention outcomes of i. The pre-intervention period is set as before the initial "softer" ban on open burning in Northern Thailand, in 2013. The donor pool consists of provinces in the rest of Thailand, districts in Myanmar, and provinces in Laos that were never subject to a similar ban. This requires assuming that the donor units did not experience a major policy intervention that would alter their outcomes. Although some of the donor regions took actions against open burning during the study period, to the best of my knowledge, they did not have a policy as strict or comprehensive as the one in Northern Thailand. In Myanmar, the main policy tools used to combat fires appear to have been attempts to raise public awareness, and inspection of some factories (Ministry of Information of Myanmar, 2022). In Laos, a wider regulatory framework was established to support public awareness campaigns and the allocation of responsibilities for prevention and control to local authorities. However, it suffered from limited resources for monitoring and control, especially in remote areas (KPL Lao News Agency, 2015; Phrasisombath, 2019).

For each i in the $N_{treated} = 9$ treated provinces, I find a vector of non-negative weights $W_i = (w_{i1}, \ldots, w_{iN_c})$ that sums to one. These weights are assigned to the N_c donor units. The penalized synthetic control method chooses these weights by solving the following optimization problem:

$$\min_{W_i} \|X_i - \sum_{j=1}^{N_c} w_{ij} X_j\|_V^2 + \lambda \sum_{j=1}^{N_c} w_{ij} \|X_i - X_j\|_V^2$$

Subject to

$$w_{ij} \geq 0, \quad \text{and} \quad \sum_{j=1}^{N_c} w_{ij} = 1$$

The first term in the minimization problem is based on the "classic" synthetic control method, and ensures that the synthetic unit closely matches the treated unit's pre-treatment characteristics X_i. The second term regularizes the weights by penalizing contributions from donor units that are dissimilar to the treated unit, using the parameter $\lambda \geq 0$. This guarantees a unique solution for each treated unit and reduces extrapolation bias, as synthetic controls composed of donor units that are individually very different from the treated unit are less likely. When $\lambda = 0$, the problem reduces to the standard synthetic control method, while when $\lambda \to \infty$, it converges towards nearest-neighbour matching (Abadie & L'Hour, 2021).

With the optimal weights $w_{ij}(\lambda)$, the counterfactual for each treated unit is estimated as $\hat{Y}_{i,t}^0 = \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{j,t}$, and the average treatment effect (ATE) at time t is:

$$\hat{\tau}_t(\lambda) = \frac{1}{N_{treated}} \sum_{i=1}^{N_{treated}} \left(Y_{i,t}^1 - \hat{Y}_{i,t}^0 \right)$$

The statistical significance is assessed using permutation inference, following Abadie and L'Hour (2021). I generate a reference distribution of placebo effects under "random" assignment of the treatment. This is done by iteratively applying the synthetic control method to each donor unit, treating it as a placebo case. The test statistic, $\hat{T}$, is the ratio of the mean squared prediction errors (MSPEs) after and before the treatment. This accounts for differences in pre-treatment fit across units:

$$\hat{T} = \frac{MSPE_{\text{post}}}{MSPE_{\text{pre}}} = \frac{\frac{1}{|\mathcal{T}_1|N_{treated}} \sum_{t \in \mathcal{T}_1} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}^0_{i,t})^2}{\frac{1}{|\mathcal{T}_0|N_{treated}} \sum_{t \in \mathcal{T}_0} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}^0_{i,t})^2}$$

The null hypothesis is that the treatment does not have an effect, implying that the test statistic $\hat{T}$ for the treated units should not be unusually large compared to the distribution of the placebo statistics. The p-value is the proportion of placebo statistics that is at least as large as the statistic for the treated group.

My main estimation evaluates the effect of the ban on the number of fires in Northern Thailand in the first five months of the year, when most of the fires are set. The outcome variable of the main specification is fire density (fires per 100 km^2), normalized by each unit's area.[3] The weights of the SCM are calculated using data from $t2013$, prior to the start of the ban on open burning in Northern Thailand.[4] The variables used are fire density, climatic conditions, average elevation, and land uses (share of urban, forest and cropland areas) in each unit. I estimate the ATE of the ban in various periods across all treated provinces for two enforcement "regimes" - a "weak enforcement" regime (2013-2016) and a "strict enforcement" regime (2017-2020). The ATE for a given enforcement period $\mathcal{T}_p$ is:

$$\hat{\tau}^{(p)} = \frac{1}{N_{treated}|\mathcal{T}_p|} \sum_{i=1}^{N_{treated}} \sum_{t \in \mathcal{T}_p} \left(Y_{it} - \hat{Y}^0_{i,t} \right)$$

In addition to the main estimation, the ban may have had a displacement effect, leading to an increase in fires during some periods of the year over others. To evaluate the potential displacement of fires, I construct a separate synthetic control for specific periods of the year, similarly to Nguyen Huy et al. (2024). For example, to evaluate the effect of the ban on fires in the month of March, I use pre-2013 covariates and the number of fires from March only, across all units, to estimate a March-specific synthetic control for each treated unit. Similarly, to evaluate heterogeneity, I construct unique synthetic controls for each outcome

[3] As some larger units have very high levels of fires, normalizing by units' areas mitigates the risks of units reaching extreme values that cannot be reproduced by synthetic units (Abadie, 2021).

[4] Synthetic controls are estimated using a modified version of the R package `pensynth` (van Kesteren & Slaughter, 2024), which implements the penalized synthetic control estimator from Abadie and L'Hour (2021).

variable. As there are separate synthetic controls for each outcome variable and for each period, the estimates for each period do not sum up to the aggregate estimate.

I also evaluate the effect of each enforcement regime for each half-month period, denoted as h. This allows to evaluate the ban's immediate impact and displacement effects, by exploiting the variation in treatment timing within each year (differences in the dates of ban set by each province). Thus, I estimate treatment effects separately for provinces currently under a ban ($U_{h,y}$), about to enter a ban in the next half-month ($J_{h,y}$), or having just exited a ban ($A_{h,y}$). For each enforcement period $\mathcal{T}_p$, the treatment effect $\hat{\tau}_k$ for category $k \in \{U, J, A\}$ is computed as a weighted average of the period-specific estimates $\hat{\tau_{k,h,y}}$, with the weights corresponding to the number of treated units $n_{k,h,y}$ contributing to each estimate:

$$\hat{\tau}_k = \frac{\sum_{h,y} n_{k,h,y} \cdot \hat{\tau}_{k,h,y}}{\sum_{h,y} n_{k,h,y}}$$

Similarly, the p-values $p_{k,h}$ for statistical inference are computed using placebo tests for donor units in each (h, y) period, and aggregated using the same weights as a weighted average:

$$\hat{p}_k = \frac{\sum_{h,y} n_{k,h,y} \cdot p_{k,h,y}}{\sum_{h,y} n_{k,h,y}}$$

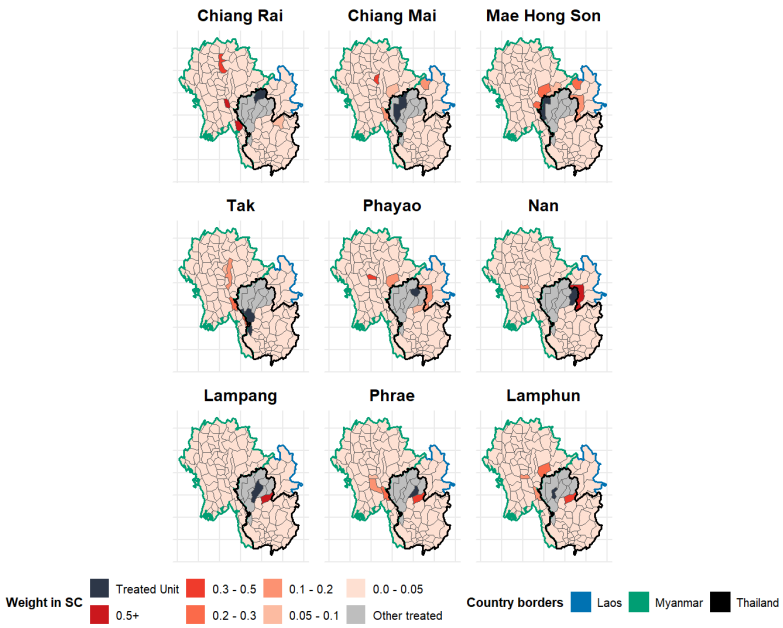
5 Results

5.1 Aggregate effects

My main estimation looks at the effect of the ban on the total number of fires detected by MODIS throughout and around the dry season, from January to May. The synthetic controls for this specification are constructed using the weights shown in Table 12, and visualised in Figure 3. The synthetic controls weights are dominated by provinces in northern and central Thailand, eastern Myanmar, and two bordering provinces in Laos. This is unsurprising, as these are mostly the provinces located in proximity to the treated ones. This estimation (and all of the following estimations) look at the effect on the number of fires detected by MODIS. Thus, estimated negative effects could represent "real" reductions in the incidence of fires, driven by fewer fires set by residents of Northern Thailand. However, it could also reflect an increase in fire evasion attempts. I will discuss this in more detail in subsection 5.5.

The synthetic control for the January-to-May dry season, shown in Figure 4, closely tracks the actual number of fires before the initial 'soft' ban in 2013, validating the counterfactual. After 2013 and particularly after the stricter enforcement began in 2017, the lines diverge

Figure 3: Synthetic control weights. To improve visibility, provinces in Southern Thailand, Eastern Laos and North-West Myanmar are not shown in the maps. The full map is shown in Figure 11.



for some of the provinces. As shown in Table 2, the ban led to a reduction of 3.24 fires per 100 km^2 from 2017-2020, with the result being significant at the 5% level. As the total area of the nine treated provinces is around 105,200 km^2 (1,052 units of 100 km^2, such that all results could be multiplied by 1,052 to obtain the effect in terms of number of fires), these results correspond to 3,419 fewer fires per year across the nine treated provinces. These effects correspond to a reduction of 28% in the number of detected fires compared to the synthetic version during the period with stricter enforcement. In contrast, the ban had no statistically significant effect during the weak enforcement period (2013-2016), suggesting that the policy's effectiveness was conditional on actual enforcement.

The effects are then examined for each month from January to May separately, and shown in the different rows in Table 2. As explained earlier, separate synthetic controls are constructed for each treated province in each period.[5] The decrease in fires throughout the dry season appears to be driven by a reduction in detected fires during March. During the four year-period of stricter enforcement, every March saw a reduction of 2.42 fires per 100 km^2 around Northern Thailand compared to the synthetic units (corresponding to a reduction of 20.6%), significant at the 10% level. In May, meanwhile, fires are estimated to have increased in years with seasonal ban and strict enforcement. I estimate that every May between 2017-2020, there were, respectively, 0.29 more fires per 100 km^2 in the treated units, with the result

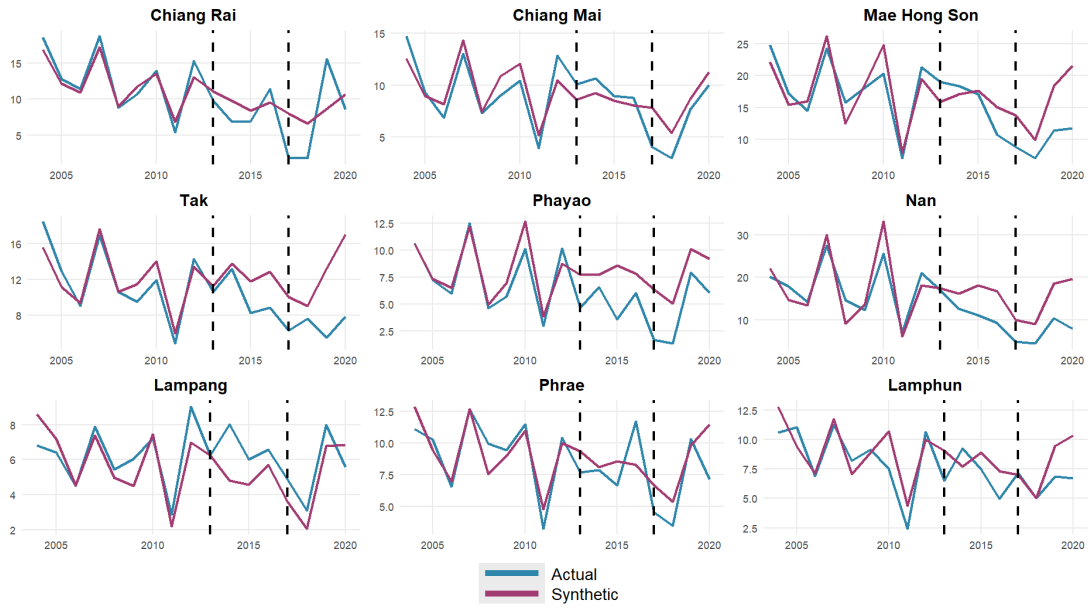
[5]This implies that the weights of donor units in the synthetic regions differ over time, and that the estimated effect in each period does not sum to the total estimated effect.

Table 2: Effect on *total fires*: change per 100 km^2 and percentage change

Period	Strict enforcement (2017–2020) (1)	Weak enforcement (2013–2016) (2)
Jan–May	-3.24^{*}	-1.06
	(0.08)	(0.30)
	[-28.0%]	[-7.2%]
January	-0.07	-0.06
	(0.45)	(0.53)
	[37.4%]	[25.9%]
February	-0.62	-0.24
	(0.19)	(0.44)
	[-17.4%]	[-13.0%]
March	-2.42^{*}	-0.50
	(0.08)	(0.21)
	[-20.6%]	[5.9%]
April	0.18	0.56
	(0.14)	(0.25)
	[88.0%]	[79.3%]
May	0.29^{**}	0.05
	(0.01)	(0.16)
	[373.8%]	[97.4%]

The table reports the absolute change in number of total fires per 100 km^2 in an administrative. The values reflect the average yearly difference between the nine treated units and their synthetic controls. The effect in percentages is in squared brackets. The p-values are determined using a permutation inference. MODIS fires above the 30% confidence threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4: Synthetic control and actual fire values in January to May every year by province. The vertical lines indicate 2013 (the beginning of the soft ban) and 2017 (the beginning of the stricter ban).



being significant at the 5% level. Figure 17 shows the effect of the ban on fires in May, showing clear increases in fires in May during the period with stricter enforcement in some of the provinces. The increase in May is relatively small in absolute magnitude and far below the reduction in March, such that the aggregate effect is large and negative. Still, the effects in May represent a large increase of 373% more fires in the average unit in May compared to the synthetic units. These results suggest that the stricter enforcement of the seasonal ban pushes some villagers to postpone their planned burning of crops or forests to after the ban, when burning is allowed again, thereby extending the length of the burning season.

5.2 Heterogeneity

To explore heterogeneity, I estimate the effect of the ban for various outcomes. To do so, I construct a separate synthetic control for each unit, for each period and outcome of interest, and estimate the average treatment effect separately for each period-outcome combination. I begin by examining the timing of the fires within the day. I distinct between day fires (detected by the Terra satellite at 10:30 and Aqua at 13:30), and night fires (detected by Terra at 22:30 and Aqua at 1:30). Results for the strict enforcement period are shown in Columns (4) for daytime fires and (5) for nighttime fires in Table 3. Daytime fires - which constitute most fires - drive the aggregate results, decreasing overall and in March while increasing in May Meanwhile, there is no estimated effect for nighttime fires on aggregate,

while they are estimated to increase in April.[6] This result contrasts with that of Cao and Ma (2023), who find a decrease only in nighttime fires under a (non-seasonal) fire ban in China. This might result from different sources of monitoring. While Cao and Ma (2023) suggest that monitoring of the ban in China was mainly done by patrols, the Thai authorities also used remote sensing to monitor fires. As MODIS is better at detecting fires during the day than at night, remote sensing makes daytime fires easier to detect, decreasing the relative costs of setting fires during the night. Next, I compute separately the effects for fires detected by the Aqua satellite and for fires detected by the Terra satellite. The effects for the Aqua, shown in Column (2) of Table 3, mirror that of the total fires, decreasing in the stricter enforcement period overall and particularly in March, while increasing in May. Meanwhile, fires detected by the Terra satellite (shown in Column (3)) do not increase throughout the dry season, but increase in April and May. When examining specifically fires detected by each overpass of each satellite (Table 11), I find that the results are mostly driven by the effect on fires detected by Aqua's daytime overpass at 13:30. This overpass is the one when most fires are detected, and during the strict enforcement period of the seasonal ban it sees a large reduction in fires. There is also a small decline in the number of fires detected by Aqua's nighttime overpass. Meanwhile, the number of fires detected in each of Terra's overpasses does not change in the five months period in the strict enforcement period. These results suggest that most of the reduction documented in the main estimation is driven by fewer fire detection by Aqua, in particular in its daytime overpass.

I proceed by computing the effects for fires detected by the VIIRS sensor abroad the S-NPP satellite. As VIIRS has better spatial resolution (375m compared to 1 km of MODIS), it could be interpreted as a proxy for smaller fires. This estimation uses weights constructed from the MODIS data, which introduces a potential for bias since these weights may not accurately represent the pre-treatment trends of VIIRS fires.[7] The estimated treatment effects for fires detected by VIIRS, shown in Table 13 are statistically indistinguishable from zero.

I then compute the effects depending on the land use in which the fires are taking place, shown in Table 7. The seasonal bans had no effects on fires taking place in croplands, while forest fires exhibit similar behaviour as the total number of fires, with a reduction of 2.81 fires per 100 km^2 throughout January to May. While these results suggest that farmers do not comply with the ban, this result may be affected by inaccurate land classification, with small agricultural plots falling into pixels that are largely forested, such that they are

[6] Similarly to the estimations by period, the estimated effects on daytime and nighttime fires do not sum to the overall effect due to unique synthetic controls used for each estimation

[7] I cannot compute separately synthetic units based on VIIRS fires, as VIIRS is available only from late January 2012. In cases of a very short pre-treatment period, synthetic controls should not be used as they would suffer from high levels of bias (Abadie, 2021).

Table 3: Effect on fire outcomes during strict enforcement (2017–2020)

Period	Total fires	Satellite		Time	
		Aqua	Terra	Day	Night
	(1)	(2)	(3)	(4)	(5)
Jan–May	-3.24^{*}	-3.32^{**}	-0.16	-3.75^{**}	0.01
	(0.08)	(0.04)	(0.18)	(0.02)	(0.05)
	[-28.0%]	[-40.6%]	[-0.5%]	[-34.1%]	[24.9%]
January	-0.07	-0.02	-0.03	-0.05	-0.00
	(0.45)	(0.49)	(0.34)	(0.49)	(0.24)
	[37.4%]	[23.1%]	[27.9%]	[-1.4%]	[304.1%]
February	-0.62	-0.53	-0.11	-0.54	-0.05
	(0.19)	(0.22)	(0.19)	(0.22)	(0.19)
	[-17.4%]	[-22.4%]	[-13.2%]	[-21.9%]	[340.6%]
March	-2.42^{*}	-2.14^{*}	-0.29	-2.39^{*}	0.03
	(0.08)	(0.07)	(0.16)	(0.06)	(0.13)
	[-20.6%]	[-23.7%]	[-2.0%]	[-24.2%]	[172.2%]
April	0.18	-0.22	0.36^{*}	-0.27	0.16^{*}
	(0.14)	(0.14)	(0.08)	(0.13)	(0.07)
	[88.0%]	[6.6%]	[44936.2%]	[2.5%]	[192.3%]
May	0.29^{**}	0.19^{**}	0.07^{**}	0.24^{**}	0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.17)
	[373.8%]	[381.6%]	[432.7%]	[235.2%]	[924.1%]

This table reports the change in number of fires detected per 100 km^2 in an administrative unit from 2017 to 2020. Percentage changes are shown in square brackets. Night fires refer to fires detected during nighttime overpass (Terra at 22:30 and Aqua at 1:30), and day fires to ones detected during daytime overpass (Terra at 10:30 and Aqua at 13:30). Aqua and Terra fires refer to fires detected by each satellite. MODIS fires above the 30% confidence threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

classified as forest fires. Looking at fires taking place within areas with different levels of crop production, I find that fires were only reduced in areas with the lowest quantile of crop production (see Table 14). These areas likely reflect very small agricultural production, consistent with the limited effect estimated in croplands. There are no clear differences when examining separately fires taking place in areas where the dominant crop is rice, compared to ones where the dominant ones are cassava or maize (Table 15). I then examine fire hotspots depending on their fire radiative power (FRP), a measure of the intensity of a fire. The results are shown in Table 16. I split the sample of fires into high and low intensity fires, classified depending on whether their radiative power is above or below the median in the entire sample. Both change in a similar way following the ban, with the effects for high intensity fires being larger in magnitude, suggesting that the effects of the main specification are largely associated with changes in high intensity fires.

Without a direct measure of enforcement, I test whether the effect of the ban was larger for fires taking place in the proximity of roads and urban centres, where enforcement might be easier. The results are shown in Table 8 and 9, respectively. The effects go in the same direction for fires above the median distance from urban centres and roads within the same country or province. However, the estimated effects for fires with above median distances to both urban centres and roads are larger, contrasting the idea that compliance would be higher where enforcement costs are likely lower. Instead, these results are consistent with the previous set of results, that suggest that the effect of the ban is largely driven by high intensity forest fires, which tend to be further away from urban centres. Finally, I examine the effects depending on the province, rather than aggregating the provincial results into an average treatment effect. The results, shown in Table 10, suggest that the provinces that drive the aggregate effects are Mae Hong Son and Tak, two large provinces with sizeable forest area that border Myanmar. Meanwhile, the estimated effects for richer provinces like Lamphun and Chiang Mai, that likely have a stronger administrative capacity, are statistically indistinguishable from zero.[8]

5.3 Robustness

I run a series of tests to examine the robustness of my results, shown in Table 17. First, the same fire could be detected more than once in a given day, or in consecutive days, or, if the same fire persists. I therefore follow Balboni et al. (Forthcoming) and Cao and Ma (2023) in treating fire hotspots within 1km^2 in the same day or in consecutive days as the same

[8]Note that while the estimated effect for Chiang Rai is negative and significant at the 10% level, it masks heterogeneity. As shown in Figure 4, Chiang Rai initially saw a reduction in fires compared to the synthetic version, but later an increase. The aggregate estimated effect, which accounts for all four years, is statistically significant as the effect in all years was large in that province.

fire, which reduces the average number of fires per 100 km^2 from 7.76 to 6.94. The results are largely unchanged, as shown in Column (3). I also test whether the effects are linked to pixels that are falsely classified as fires, by only including fire spots with above the 75% confidence threshold in the MODIS dataset. Results are shown in Column (4) and retain the same direction, although the absolute value is lower due to the lower number of fires.

I test whether the results are robust to changes in the synthetic control generation. First, I trim the pre-treatment outcomes period to start at a later year. Instead of using 2004-2012 as the pre-treatment years, I use 2008-2012. Column (2) shows the results, that remain similar to those of the main specification, suggesting that the results are not driven by weights chosen largely based on years far before the beginning of the treatment. I also test whether my results are robust to the exclusion of certain donor units. If some other regions had their own policies which effected the occurrences of fires, this may generate bias in my results. My results are robust to excluding donor units from Myanmar (Column (5)), from Laos (Column (6)), and from Thailand (Column (7)). I also evaluate whether the results are sensitive to changes in the set of outcomes used to generate the donor units' weights. The results are robust to excluding the climatic variables, as shown in Column (8).

Another concern is that the period- and outcome-specific synthetic controls may lead to over-fitting. Hence, I estimate the effects for all periods while fixing the synthetic controls based on the weights determined in the full January to May period. The results are shown in Column (9). The estimated effect using this method for the period from January to May is, by design, identical to the baseline estimation of Column (1). The estimated effects in May are robust to fixing the set of weights, remaining positive and statistically significant, while the estimated effects in March is no longer statistically significant. The different results for March likely result from this estimation failing to capture March-specific characteristics for the treated units. While it is more computationally efficient and reduces over-fitting, it does not account for the timing of the burning season in different provinces.

Finally, I perform a series of backdating tests (Abadie, 2021). This involves constructing the synthetic controls using a truncated pre-intervention period which ends at least a year before the actual start of the treatment. Then, I can evaluate the predictive accuracy, thereby testing the out-of-sample prediction, and whether there is a null effect just before the treatment. Additionally, it allows testing the robustness of the results to the truncated pre-intervention period. I run backdating tests with the the last year used for generating the synthetic controls being 2009, 2010 and 2011, respectively. As shown in Table 18, and in Figures 8 through 10, there is no statistically significant effect for the "treatment" in the validation periods. This provides strong evidence that the model is capturing the pre-treatment period well, and that the results are not driven by pre-existing trends. The main

findings, of an overall reduction in fires during the strict enforcement period and a small increase in May, remain robust across all backdating tests.

5.4 Effect of being under the ban

I then estimate the effects of being under the ban, just before it, or just after it. As explained in Section 4, I use half-month-specific synthetic controls for this analysis, which allows me to exploit the variation in treatment timing. The results, shown in Table 4, suggest that while the point estimate for being under the ban is negative, it is not statistically distinguishable from zero. This indicates that a ban, by itself, may not be enough to deter burning without a longer-term effect. However, the results show a significant increase in fires just after the ban ends, but only during the stricter enforcement period. I estimate an increase of 1.77 fires per 100 km^2 in the half-month immediately following the ban, which corresponds to 1,862 more fires across the nine treated provinces in two weeks. This is consistent with the temporal displacement of fires found in the main specification. Meanwhile, the period with weak enforcement shows no such statistically significant increase after the ban ends.

A potential limitation of this method is the pooling of different ban periods. The reported effects are weighted averages of half-month-specific estimates, where the weights are determined by the number of units under each treatment status. For example, $n = 161$ represents 161 half-weeks weeks when treated units were under the ban over the four-year period, averaging an estimated 2.24 months per year. This aggregation may indistinct heterogeneity in enforcement, as it pools together most two-months long bans, with a few longer bans (such as the nearly four-months long ban in the province of Chiang Mai in 2020) where the enforcement might be weaker over time.

6 Estimating detection avoidance

The estimated reduction in fire detections following a burning ban can be caused by two primary mechanisms. First, it could reflect compliance with the moratorium, reflecting a genuine reduction in burning activity. Second, it could be driven by strategic evasion by farmers, who shift burning times to avoid detection by FIRMS satellites, that are some of the tools used by the Thai authorities to monitor fires (Smith, 2024). MODIS detects fires using the Aqua and Terra satellites, which pass over Northern Thailand at approximately 1:30 and 10:30 (AM and PM), respectively. Farmers can evade monitoring by burning after Aqua's overpass and ensuring that the fire extinguishes before Terra's pass, roughly nine hours later.[9] Qualitative evidence suggests that such evasion occurs among farmers

[9]The S-NPP satellite, equipped with VIIRS, was monitoring fires during the period of the ban as well. Its overpass times are roughly identical to those of Aqua, at 1:30 (NASA FIRMS, 2023b). Therefore, while

Table 4: Weighted treatment effects on total fires by ban status and enforcement period

Ban Status Period	Stricter enforcement (2017-2020)	Weak enforcement (2015-16)
During	−6.44	−1.68
	(0.16)	(0.26)
	[n=161]	[n=67]
Just after	1.77*	1.46
	(0.08)	(0.14)
	[n=35]	[n=14]
Just before	−0.772	−0.182
	(0.28)	(0.43)
	[n=34]	[n=14]

This table shows weighted treatment effects on total fires across different ban status periods and enforcement regimes.
p-values in brackets. Sample sizes (number of half-months under which treated units were known to be in each treatment category over the entire period) in squared brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

(Mostafanezhad & Evrard, 2021), although its scope remains unclear. Figure 5 shows the daily overpass times of Aqua and Terra and the nine-hour "evasion windows" that exist between them. Although some fires, particularly agricultural fires, can extinguish within just 2–3 hours, other fires can burn for much longer. Thus, fires that are purposefully set after Aqua's overpass to avoid monitoring, may or may not extinguish before Terra's pass and be detected by Terra. In this section, I attempt to quantify the degree to which evasion explains the observed reduction in fire detections.

Let F_s^t be the number of fires detected by sensor $s \in$ (Aqua, Terra) at overpass $t \in (D, N)$, with D representing the daytime overpass, and N the nighttime overpass. To examine shifting, I focus on F_{Aqua}^D, fires detected by Aqua during the preceding daytime pass (13:30) that are shifted to the afternoon. As very few fires are detected in Aqua's nighttime pass due to MODIS's limitation in detecting nighttime fires, examining the relationship between F_{Aqua}^N (1:30) and F_{Terra}^D (10:30) is unlikely to uncover any evasion attempts. Moreover, a shift in burning time to the afternoon might be reflected in Aqua's nighttime fires, as some fires are likely to still burn 12 hours after being set. Thus, rather than estimating the extent of evasion out of the full reduction in fires, I examine its scope out of the reduction estimated during the Aqua daytime pass, which is shifted into the later passes and in particular into Terra's nighttime overpass (F_{Terra}^N). This likely represents some underestimation of the full extent of evasion, but given that the majority of fires are captured during Aqua's daily pass, the scope of this underestimation is likely small.

it is better at detecting small fires and nighttime fires, VIIRS is unlikely to capture a fire that was shifted to the window between Aqua and Terra and extinguished by Terra's overpass.

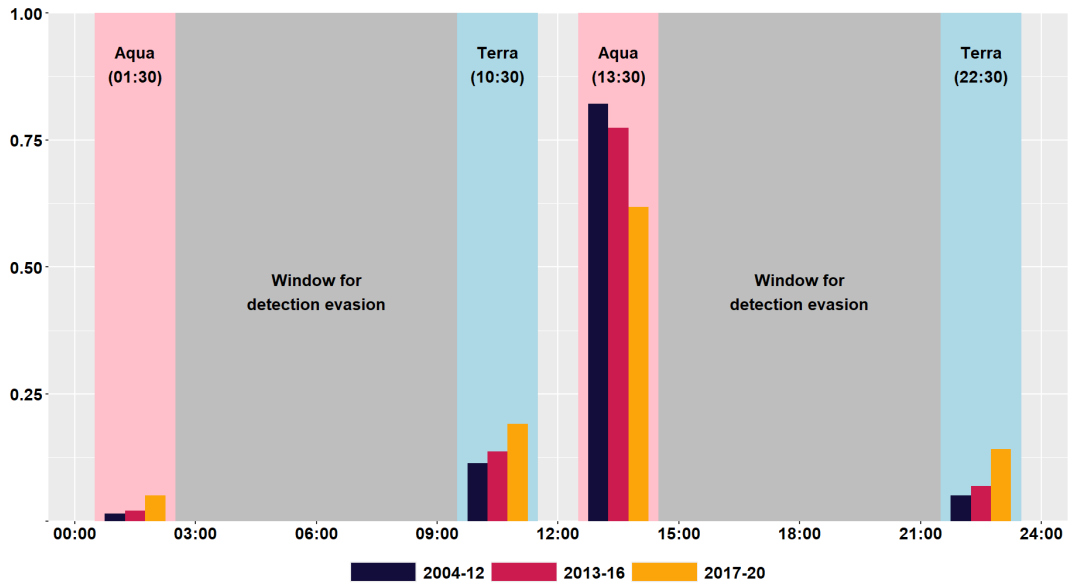


Figure 5: Satellite Overpass Times and Evasion Windows. The figure illustrates the daily overpass times for the Aqua and Terra satellites over Northern Thailand and the share of fires detected at each overpass. Farmers can evade detection by burning in the ∼9-hour windows between satellite passes.

I compare the estimated reduction in fires detected in Aqua's daytime pass under the ban, denoted as ΔF^D_{Aqua} and shown in Column (4) of Table 11, with what we could expect if farmers do not alter their burning timing, $\Delta \widehat{F^D_{Aqua}}$. I assume that in the absence of evasion from monitoring, a reduction in detected fires in each satellite pass should be proportionate to the pre-ban share of fires detected during that overpass. This assumes that the duration and extinguishment rate of fires does not change systematically in response to the ban. This is a strong assumption which may lead to underestimation of the true extent of detection evasion, but is attributed to the lack of data in the FIRMS dataset regarding the extinguishing time of fires. The shares of fires detected prior to the ban by Terra during its nighttime pass, and by Aqua during its daytime pass, out of the combined output, are:

$$s^D_{Aqua} = \frac{F^D_{Aqua}}{F^D_{Aqua} + F^N_{Aqua} + F^D_{Terra} + F^N_{Terra}}$$

Under the assumption of no evasion from monitoring, farmers should not change the burning times. Hence, s^D_{Aqua} should remain constant over time. The counterfactual $\widehat{\Delta F^t_s}$ is computed by multiplying the shares s^D_{Aqua} with the combined estimated reduction, $\Delta F_{reduced} = \sum_{s,t} \min(\Delta F^t_s, 0)$, where I sum all negative effects across all satellite passes:

$$\Delta \widehat{F^D_{Aqua}} = s^D_{Aqua} \times \Delta F_{reduced}$$

The gap between the expected reduction in fires detected during Aqua's daytime pass and the actual reduction, is denoted as ρ:

$$\rho = \Delta F^{D}_{Aqua} - \Delta \widehat{F^{D}_{Aqua}}$$

$\rho \neq 0$ is interpreted as indicating a change in the propensity of farmers to burn at specific times due to the ban. Since $\Delta F^{D}_{Aqua} < 0$ in the main specification, $\rho < 0$ implies that the reduction in Aqua's daytime detections is larger than expected ($|\Delta F^{D}_{Aqua}| > |\Delta \widehat{F^{D}_{Aqua}}|$), while the other satellite's overpasses' detections decrease by less than expected or increase. This is suggestive of evasion from monitoring by satellites, with more fires than expected still burning by Terra's nighttime pass. This is consistent with temporal shifting of fires into Terra's detection window, and may capture evasion attempts. $\rho > 0$, meanwhile, implies that Aqua's daytime pass detects more fires than expected. This can be interpreted as evasion from monitoring by patrols, rather than by satellites. If farmers intend to avoid detection by patrols of fire control units, they should prefer to burn during the day than at night, as detecting a fire from a distance with the naked eye is easier during the night (Cao & Ma, 2023). Thus, a shift to daytime fires is suggestive of avoiding patrols' monitoring. In such a case, the actual reduction in Aqua's daytime fires is smaller than expected, such that $|\Delta F^{D}_{Aqua}| < |\Delta \widehat{F^{D}_{Aqua}}|$ and $\rho > 0$. I take ρ, the gap between the actual number of fires to the expected one in the absence of shifting, and divide it by the total estimated reduction in fires, $\Delta F_{reduced}$. As I am only interested in evasion from satellites, I take the maximum of this term with 0, to ensure that the estimated share of reduction attributed to evasion will not be negative:

$$\gamma^{D}_{Aqua} = \max(0, \frac{\rho}{\Delta F_{reduced}})$$

γ^{D}_{Aqua} represents the proportion of the observed reduction in fires under the ban, that can be attributed to temporal shifting. The results for different periods are shown in Table 5. Given the conservative assumptions I put forth, γ^{D}_{Aqua} could be interpreted as a lower bound. Overall, throughout January to May, at least 10.6% of the estimated reduction in fires is estimated to be caused by detection avoidance as documented in shift of fires out of Aqua's daytime overpass at 13:30 into the afternoon window. At January and May, when none of the regions are under ban, there is no estimated detection avoidance, which is consistent with the theoretical expectation. In February, March and April, the estimates for detection avoidance out of the reduction range between 6.3% and 14.8%.

Theoretically, the upper bound of the extent of evasion out of the observed reduction could be the entirety of the reduction. However, this is very unlikely. The estimated increase in fires following the end of the ban likely represents a displacement of burning from the

Table 5: Detection Avoidance Estimates during Strict Enforcement (2017–2020)

	(1)	(2)	(3)	(4)	(5)	(6)
Period	s^D_{Aqua}	$\Delta F_{reduced}$	$\Delta \widehat{F^D_{Aqua}}$	ΔF^D_{Aqua}	ρ	γ^D_{Aqua}
Jan-May	0.813	−3.803	−3.091	−3.497	−0.406	0.106
January	0.797	−0.049	−0.039	−0.026	0.013	0.000
February	0.795	−0.569	−0.452	−0.488	−0.036	0.063
March	0.835	−2.394	−1.999	−2.179	−0.180	0.075
April	0.852	−0.303	−0.258	−0.303	−0.045	0.148
May	0.735	0.000	0.000	0.185	0.185	0.000

Column (1) shows the share of fires detected by Aqua's daily overpass out of all fires in Northern Thailand, prior to the ban. Column (2) shows the sum of all negative effects estimated in Table 11. Column (3) shows the expected theoretical reduction on Aqua's daytime detection, by multiplying the value of Column (1) with Column (2). Column (4) shows the actual reduction, as shown in Column (4) of Table 11. Column (5) shows the difference between the actual reduction in Column (4) and the counterfactual reduction in Column (3). Column (6) represents the gap from Column (5), divided by the combined reduction from Column (2).

period of the ban, and not additional burning. Thus, there is some actual reduction in fires during the ban, such that the observed reduction is not fully explained by evasion attempts. Furthermore, for evasion to account for the entirety of the reduction, all fires must extinguish within nine hours. This is very unlikely, especially as the window is likely somewhat shorter than nine hours. This is due to the uncertainty about the exact timing of Aqua's overpass each day, which is around but not precisely at 13:30, such that villagers may not burn immediately after Aqua's overpass. Nevertheless, the actual decrease in the incidence of fires due to the moratorium is lower than estimated in Section 5, due to the limited temporal resolution of the monitoring satellites and behavioural response.

7 Conclusion

I present causal evidence on the effect of a seasonal burning ban in Northern Thailand. I find that the effect of the ban in the nine treated provinces was conditional on its enforcement, with no impact during the weak enforcement period between 2013 and 2016. When enforcement was stricter, between 2017-2020, the number of fires detected by the MODIS sensor from January to May decreased by around 28%. However, this reduction was followed by a temporal displacement of burning activity. The burning season was extended, with more fires detected annually after the bans had ended, around the end of April and May. The effects were driven by high-intensity fires, taking place in relatively remote forested areas with low agricultural output, and detected during the day. These finding may point to a key benefit

of using remote sensing in lower-income settings: its ability to increase enforcement capacity in hard-to-reach areas. My results could be explained by the use of satellites for monitoring in Thailand. Satellites enable cheaper, quicker, and more accurate monitoring of fires in remote locations, especially during the day. Although there could be other explanations for my results, this points to the potential benefit of using remote sensing in lower-income settings to increase enforcement capacity.

The analysis also reveals a limitation of satellite monitoring. Villagers responded to the predictable overpass times of the monitoring satellites, by adjusting their burning schedules to avoid detection. I estimate that at least 10.6% of the observed reduction in fires can be attributed to this detection avoidance. This highlights a behavioural response to satellite-aided monitoring of regulatory violations. The effectiveness of such systems can be undermined when the illegal activity is short-lived, and when the monitoring tools have limited and predictable temporal and spatial resolution. Further research could improve the understanding the scope and characteristics of detection avoidance, and support policies that could tackle these behavioural responses.

My work only looks at the effects on the number of detected fires, with data limitations preventing an evaluation of the policy's welfare consequences. Future work could address this gap by examining how the reduction and displacement of burning activities alter outcomes such as air pollution or health. These effects may be large, as shifting the timing of burning even within the same day potentially having substantial health impacts (Lan et al., 2022).

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Appendix

A. Synthetic control method

Traditional synthetic control method

The synthetic control method constructs a counterfactual for a treated unit with a weighted combination of the untreated "donor" units, that closely resembles the treated unit prior to the treatment (Abadie, 2021). Formally, let $Y_{i,t}$ denote the observed outcome for unit i at time t. For treated units, the outcome at time t during the post intervention period $t \geq T_0 = 2013$, is $Y^1_{i,t}$, while the unobserved counterfactual outcome in the absence of the ban is $Y^0_{i,t}$. The causal effect of the treatment for unit i at time t is $\tau_{i,t} = Y^1_{i,t} - Y^0_{i,t}$. Since $Y^0_{i,t}$ is not directly observable, the synthetic control method approximates it by constructing a weighted average of the donor units. The synthetic control for treated unit i is constructed by solving a constrained optimization problem. There are $N_{treated}$ treated units, and N_c donor units. Let $W_i = (w_{i1}, \ldots, w_{iN_c})$ denote the vector of non-negative weights ($w_{ij} \geq 0$) assigned to each donor unit $j \in \{1, ..., N_c\}$ to construct the synthetic version of treated unit i. To avoid extrapolation, the weights must sum to 1, such that $\sum_{j=1}^{N_c} w_{ij} = 1$. A synthetic control is estimated separately for each treated unit i. The weights are chosen to minimize pre-treatment differences between the treated unit's characteristics and its synthetic version:

$$\|X_i - X_0 W_i\|_V = \sqrt{(X_i - X_0 W_i)' V (X_i - X_0 W_i)}$$

Where X_i is a matrix of pre-treatment characteristics for treated unit i, X_0 is the matrix of corresponding characteristics for the donor units, and V a matrix that assigns weights to the pre-treatment predictors depending on their importance in explaining the outcome variable $Y_{i,t}$. The matrix is chosen to minimize the mean squared prediction error (MSPE) in the pre-intervention period, such that the synthetic control closely matches the outcomes of the treated unit prior to the treatment. With the weights, the counterfactual estimate of treated unit i in the post intervention period is:

$$\hat{Y}^0_{i,t} = \sum_{j=1}^{N_c} w_{ij} Y_{j,t}$$

With the treatment effect for a single treated unit i estimated as:

$$\hat{\tau}_{i,t} = Y^1_{i,t} - \hat{Y}^0_{i,t}$$

To obtain the average treatment effect at time t, all individual treatment effects are averaged

such that:

$$\hat{\tau}_t = \frac{1}{N_{treated}} \sum_{i=1}^{N_{treated}} [Y_{i,t}^1 - \hat{Y}_{i,t}^0] = \frac{1}{N_{treated}} \sum_{i=1}^{N_{treated}} \left[Y_{i,t}^1 - \sum_{j=1}^{N_c} w_{ij} Y_{j,t} \right]$$

Penalisation

In the presence of multiple treated and donor units, constructing a synthetic control separately for each treated unit can lead to non-unique solutions: for a given treated unit, many weighted combinations of donor units could create a synthetic control. A non-sparse solution could complicate the interpretation of the synthetic control, by leading to many potential treatment effect for a given treated unit, and by preventing understanding of the units composing each synthetic control Abadie and L'Hour (2021).[10] Furthermore, when there are many donor units and some are very close to a treated unit in the covariate space, a single untreated unit may provide close matches to a treated unit, increasing the risk of over-fitting.

Abadie and L'Hour (2021) propose adding a penalisation term λ to the optimization problem from before, that regularizes the weights given to the donor units. This guarantees the uniqueness of the solution. λ presents a trade-off between minimizing the differences between the treated unit and the aggregate synthetic control (as in the classic synthetic control), and minimizing the discrepancies between the treated unit and each single donor unit in its synthetic control (which is akin to the case of matching), with the minimization problem being:

$$\min_{W_i} \|X_i - \sum_{j=1}^{N_c} w_{ij} X_j\|_V^2 + \lambda \sum_{j=1}^{N_c} w_{ij} \|X_i - X_j\|_V^2$$

Subject to $w_{ij} \geq 0$for $j \in \{1, \ldots, N_c\}$, and $\sum_{j=1}^{N_c} w_{ij} = 1$.

The first term, as before, ensures that the synthetic control matches closely the pre-treatment characteristics of the treated unit. The second term, which is penalised using λ, discourages large weights on donors that are far from the treated unit. If $\lambda = 0$, then the standard synthetic control estimator is recovered. As $\lambda \to \infty$, the estimator converges towards nearest-neighbour matching (Abadie & L'Hour, 2021). The average treatment effect in the case of penalised synthetic control is estimated as:

$$\hat{\tau}_t(\lambda) = \frac{1}{N_{treated}} \sum_{i=1}^{N_{treated}} \left(Y_{i,t}^1 - \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{j,t} \right)$$

[10] Averaging the treated units to obtain a single aggregate unit and then computing a synthetic control for it could recover an ATE, but this masks out heterogeneity in the treatment effect and increases interpolation bias (Abadie & L'Hour, 2021).

Inference

I assess the statistical significance of the estimated treatment effects using permutation inference (Abadie & L'Hour, 2021; Abadie et al., 2010). This involves constructing "placebo" synthetic controls for donor units, and comparing the aggregate estimate for the treated units to a distribution of placebo effects generated under a "random" assignment of treatment. Rather than comparing the treatment effects directly, I use an aggregate test statistic ($\hat{T}$) based on the ratio of the aggregate mean square prediction error (MSPE) in the post-intervention period to that in the pre-intervention period. The MSPE measures the average of the squared differences between the observed outcomes and their synthetic control counterfactuals. The MSPE for a set of units N over a time period $\mathcal{T}$ is defined as:

$$MSPE = \frac{\sum_{t \in \mathcal{T}} \sum_{i=1}^{N} (Y_{it} - \hat{Y}_{i,t}^{0})^2}{|\mathcal{T}|N}$$

This MSPE is computed separately for a pre-intervention period $\mathcal{T}_0$ and for the post-intervention period $\mathcal{T}_1$ (in the case of a single post-intervention period). The aggregate test statistic $\hat{T}$ is based on the ratio of MSPE after the treatment ($MSPE_{\text{post}}$) and before the treatment ($MSPE_{\text{pre}}$). This accounts for differences in the pre-treatment fit of the synthetic control across units (Abadie & L'Hour, 2021). The test statistic $\hat{T}$ which refers to the aggregate effect is:

$$\hat{T} = \frac{MSPE_{\text{post}}}{MSPE_{\text{pre}}} = \frac{\frac{1}{|\mathcal{T}_1|N_{treated}} \sum_{t \in \mathcal{T}_1} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}_{i,t}^{0})^2}{\frac{1}{|\mathcal{T}_0|N_{treated}} \sum_{t \in \mathcal{T}_0} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}_{i,t}^{0})^2}$$

The distribution of this statistic is obtained using a permutation. This involves iteratively reassigning the treatment status at random. Each donor unit j can be treated as if it was a treated unit, such that a synthetic version for each donor unit would be created using all the remaining donor units. Then a test statistic could be estimated based on the ratio of MSPE after and before the treatment. The null hypothesis is that treatment has no effect. Under this, the treatment assignment is random, and the test statistic for the units that were actually treated should be drawn from the same distribution as the placebo statistics for the donor units.

The permutation-based p-value is computed as the share of placebo test statistics that are as extreme or more extreme than that of the aggregate treated units:

$$\hat{p} = \frac{1 + \sum_{\text{placebo}=1}^{B} I\left(|\hat{T}^{(\text{placebo})}| \geq |\hat{T}^{(\text{treated})}|\right)}{1 + B}$$

Where B is the number of permutations, such that if all donor units are used as placebos, $B = N_c$.

Because the MSPE is a squared term, it is always non-negative. The test statistic $\hat{T}$ therefore measures the magnitude of the post-treatment prediction error relative to the pre-treatment error, but not its direction (i.e., whether fires increased or decreased). Thus, the aggregate effect could be significant in the presenece of heterogenous effects, with some units having a large positive effect and others large negative effect.

Aggregate effects

To evaluate the aggregate effects of the ban, I estimate the average treatment effect (ATE) and significance, separately for two distinct enforcement regimes. The first is the weak enforced period between 2013 and 2016, with the second being the period is the stricter (although imperfect) enforcement period between 2017–2020. The ATE is computed by averaging the difference between observed outcomes and synthetic controls across treated units and across years within each enforcement period. Formally, let $\mathcal{T}_{\text{weak}}$ denote the weak enforcement period (2013–2016), and $\mathcal{T}_{\text{strict}}$ the stricter period (2017–2020). Then, the average treatment effect for each period is:

$$\hat{\tau}^{(\text{weak})} = \frac{1}{N_{treated}|\mathcal{T}_{\text{weak}}|} \sum_{i=1}^{N_{treated}} \sum_{t \in \mathcal{T}_{\text{weak}}} \left(Y_{it} - \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{jt} \right)$$

$$\hat{\tau}^{(\text{strict})} = \frac{1}{N_{treated}|\mathcal{T}_{\text{strict}}|} \sum_{i=1}^{N_{treated}} \sum_{t \in \mathcal{T}_{\text{strict}}} \left(Y_{it} - \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{jt} \right)$$

These estimates compare the outcomes of the treated units and their respective synthetic controls during each enforcement regime and thereby capture the effect of the ban under each enforcement regime. To assess the statistical significance, I use the ratio of post- to pre-treatment MSPE as described above, with the post-treatment MSPE computed separately for each enforcement regime ($\mathcal{T}_{\text{weak}}$ or $\mathcal{T}_{\text{strict}}$).

Effects of being under ban

The main limitation with the previous estimations to evaluate the effect of the ban results from the temporal and regional variation in moratorium dates. This creates a pooling problem, where estimation periods combine regions under active bans with those not subject to restrictions. To address this, I exploit the structure of the ban dates, with most of them starting in the beginning or middle of months, and ending in the middle or end of months. I therefore partition each month into two half-month periods ($h = 1, 2, ..., 24$ for a year), where each half-month represents approximately two weeks. Using this estimation presents

a trade-off: it prevents the pooling of the effects of different stages of bans, but increases the risk of overfitting the synthetic controls.

Let $B_{i,h,y}$ denote the ban status (1 or 0) of province i in half-month h of year y. I classify each province i in each y into one of three treatment categories, based on the relationship with the timing of the ban. First, $U_{h,y}$), is the set of provinces i where the ban is active, so $\{i : B_{i,h,y} = 1\}$. Then, $J_{h,y}$) is the set of provinces i where the ban will start in the next half-month, such that $\{i : B_{i,h,y} = 0 \text{ and } B_{i,h+1,y} = 1\}$. Finally, $A_{h,y}$) is the set of provinces i where the ban has ended in the previous half-month, with $\{i : B_{i,h,y} = 0 \text{ and } B_{i,h-1,y} = 1\}$. For each treatment category $k \in \{U, J, A\}$, I estimate the average treatment effect in half-month h of year y as:

$$\hat{\tau}_{k,h,y} = \frac{1}{n_{k,h,y}} \sum_{i \in k_{h,y}} \left[Y^1_{i,h,y} - \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{j,h,y} \right]$$

Where $n_{k,h,y}$ is the number of treated units that actually contribute to the estimate in category k during half-month h of year y. $n_{k,h,y}|$, the effective sample size, might differ from $|k_{h,y}|$ due to incomplete information on ban dates across all provinces and years. I drop from the sample cases when the ban status $B_{i,h,y}$ of any i is unknown. As before, I compute aggregate effects based on the enforcement regime, splitting into years of weak enforcement (2015-16, as I do not have dates of bans for any province in 2013 or 2014), and years of stricter enforcement (2017-20).

For each enforcement period,, I compute an overall treatment effect $\hat{\tau}_k$ for category k by aggregating all period-specific estimates $\hat{\tau}_{k,h,y}$ across both half-months h and years y. Thus, the treatment effect is a weighted average, where the weighted are determined by number of treated units $n_{k,h,y}$ used to estimate each $\hat{\tau}_{k,h,y}$:

$$\hat{\tau}_k = \frac{\sum_{h,y} n_{k,h,y} \cdot \hat{\tau}_{k,h,y}}{\sum_{h,y} n_{k,h,y}}$$

Similarly, the aggregate p-values $p_{k,h}$ for statistical inference (computed as before, using placebo tests for all donor units in each period), are computed as a weighted average:

$$\hat{p}_k = \frac{\sum_{h,y} n_{k,h,y} \cdot p_{k,h,y}}{\sum_{h,y} n_{k,h,y}}$$

B. Dates of the burning bans in Northern Thailand

Table 6 shows the known dates of the open-burning ban in Northern Thailand, from 2015-2020. The dates were extracted from four main sources. Dates for all nine provinces except for Tak, for the years 2015-19, were published in Thai Health Center for Ethnic Groups, Marginal People and Migrant Workers (2022). To verify the dates for 2018, I use the dates published in Office of Agricultural Extension and Development (2018). Similarly, for 2019, I extract the dates from Thai Pollution Control Department (2019). For 2020, the dates were published in The Nation (2020). For the province of Tak for 2017, the dates were extracted from Tak Provincial Administrative Organization (2017), and for 2018-2020 from the aforementioned sources. Since for many of those dates I only have a single source, or two sources that might be dependent on one another, the confidence regarding the precision of the dates is somewhat limited. Another concern is that the dates change after a given announcement, for example if a preliminary date for the end of the ban is announced initially and then is modified. I cannot fully rule out this possibility, but considering that the bans are usually set for 60 days, the dates of the end of the ban seem unlikely to be modified ex-post.

Table 6: Burning ban dates by province and year, 2015-2020

Province	2015		2016		2017	
	Start	End	Start	End	Start	End
Chiang Mai	-	-	16.02	15.04	20.02	20.04
Phrae	01.02	01.04	15.02	14.04	15.02	15.04
Nan	01.02	01.04	01.02	31.03	15.02	20.04
Phayao	14.02	14.04	16.02	15.04	16.02	16.04
Tak	14.02	13.04	-	-	17.02	17.04
Lamphun	01.02	30.04	15.02	14.04	15.02	15.04
Lampang	01.02	01.04	01.02	31.03	01.02	30.04
Mae Hong Son	-	-	01.03	29.04	01.03	30.04
Chiang Rai	-	-	17.02	16.04	17.02	17.04

Province	2018		2019		2020	
	Start	End	Start	End	Start	End
Chiang Mai	01.03	20.04	01.03	30.04	10.01	30.04
Phrae	15.02	15.04	15.02	15.04	01.02	30.04
Nan	15.02	15.04	15.02	15.04	15.02	15.04
Phayao	16.02	16.04	15.02	15.04	01.03	30.04
Tak	10.02	10.04	01.03	30.04	01.03	30.04
Lamphun	20.02	20.04	01.03	30.04	01.03	30.04
Lampang	10.02	10.04	10.02	10.04	01.03	30.04
Mae Hong Son	01.03	30.04	01.03	30.04	15.03	30.04
Chiang Rai	12.02	12.04	10.02	15.04	-	-

Dates are in dd.mm format. - represents unknown dates of a ban, which could either reflect that there was no ban, or that the dates of the ban in that province in that year are unknown. **Sources:** Thai Health Center for Ethnic Groups, Marginal People and Migrant Workers (2022) - all provinces except Tak from 2015-19. Tak Provincial Administrative Organization (2017) - Tak in 2017; Office of Agricultural Extension and Development (2018) - all provinces in 2018; Thai Health Center for Ethnic Groups, Marginal People and Migrant Workers (2022) and Thai Pollution Control Department (2019) - all provinces in 2019; The Nation (2020) - all provinces in 2020.

C. Supplementary Figures and Tables

Table 7: Effect on fires by land cover type

Period	Forest		Cropland	
	Strict (2017-2020)	Weak (2013-2016)	Strict (2017-2020)	Weak (2013-2016)
	(1)	(2)	(3)	(4)
Jan–May	-2.81^{**}	-0.80	-0.23	-0.16
	(0.05)	(0.27)	(0.79)	(0.77)
January	0.08	0.07	-0.04	-0.06
	(0.32)	(0.72)	(0.89)	(0.80)
February	-0.04	0.14	-0.07	-0.04
	(0.52)	(0.61)	(0.83)	(0.87)
March	-1.50	0.39	-0.11	-0.04
	(0.16)	(0.40)	(0.28)	(0.42)
April	-1.49	-1.37	-0.02	-0.02
	(0.22)	(0.52)	(0.37)	(0.57)
May	0.14^{**}	-0.02	0.00	-0.00
	(0.03)	(0.18)	(0.33)	(0.77)

This table estimates the change in number of fires per 100 km^2 in an administrative unit by land cover type following the ban. Forest fires occur where the majority of pixels within 1km of the fire is classified as forests; cropland fires occur where the majority of pixels within 1km is classified as cropland. MODIS fires above the 30% confidence threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Effect on fire outcomes during strong enforcement: distance from roads

Period	Province		Country	
	Below median	Above median	Below median	Above median
Jan–May	-1.74*	-1.85*	-1.44**	-1.99*
	(0.05)	(0.05)	(0.03)	(0.09)
January	-0.04	-0.01	-0.02	-0.06
	(0.36)	(0.43)	(0.49)	(0.37)
February	-0.40	-0.31	-0.34*	-0.48
	(0.21)	(0.10)	(0.08)	(0.18)
March	-1.08*	-1.29*	-0.63	-1.34
	(0.07)	(0.05)	(0.21)	(0.11)
April	-0.05	0.11*	0.01	0.06
	(0.14)	(0.08)	(0.11)	(0.15)
May	0.16**	0.12**	0.13**	0.15**
	(0.01)	(0.02)	(0.02)	(0.03)

This table reports the change in number of fires detected per 100 km^2 in an administrative unit from 2017 to 2020. Above median refers to fires taking place above the median distance from roads in the province or in the country. Paved roads are are used based on (OpenStreetMap, 2025).
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Effect on fire outcomes during strong enforcement: distance from urban centres

Period	Province		Country	
	Below median	Above median	Below median	Above median
Jan–May	-1.50*	-2.03**	-1.32**	-2.78*
	(0.07)	(0.04)	(0.05)	(0.06)
January	-0.06	-0.01	-0.07	0.04
	(0.27)	(0.63)	(0.32)	(0.42)
February	-0.20	-0.33	-0.37	-0.28
	(0.32)	(0.14)	(0.19)	(0.10)
March	-0.89	-1.56**	-0.48	-1.95*
	(0.13)	(0.05)	(0.16)	(0.08)
April	0.04*	0.12	0.09**	0.03
	(0.06)	(0.21)	(0.02)	(0.17)
May	0.12**	0.12**	0.13***	0.13**
	(0.02)	(0.04)	(0.01)	(0.05)

This table reports the change in number of fires detected per 100 km^2 in an administrative unit from 2017 to 2020. Above median refers to fires taking place above the median distance from urban centres in the province and in the country. Urban centres are defines as urban or suburban/peri-urban areas in the 2010 GHS - SMOD (Schiavina et al., 2023).
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Effect on fire outcomes during strict enforcement (2017–2020): by province

Period	**Chiang Mai**	**Chiang Rai**	**Lamphun**	**Lampang**	**Mae Hong Son**	**Nan**	**Phayao**	**Phrae**	**Tak**
Jan-May	-2.12	-1.55*	-1.52	0.56	-6.18*	-7.42	-3.44*	-1.98	-5.53***
	(0.27)	(0.01)	(0.36)	(0.52)	(0.08)	(0.14)	(0.06)	(0.13)	(0.01)
	[-29.1%]	[-22.4%]	[-15.0%]	[21.1%]	[-37.2%]	[-51.8%]	[-50.9%]	[-25.2%]	[-41.4%]
January	0.08	-0.23*	-0.34	0.11	-0.00	-0.09	-0.38	-0.07	0.28
	(0.55)	(0.07)	(0.74)	(0.70)	(0.48)	(0.40)	(0.20)	(0.75)	(0.41)
	[36.3%]	[8.5%]	[-46.0%]	[9.6%]	[411.0%]	[-32.6%]	[-64.2%]	[-36.0%]	[49.9%]
February	-0.16	-1.94	-1.07*	-0.42*	0.09	-0.55	-0.83	-0.41	-0.25
	(0.26)	(0.12)	(0.10)	(0.09)	(0.95)	(0.31)	(0.22)	(0.20)	(0.20)
	[11.3%]	[-76.3%]	[-15.4%]	[-7.5%]	[9.3%]	[-25.7%]	[-70.3%]	[-15.1%]	[33.3%]
March	-1.92	-3.54**	1.24	0.99	-4.81**	-6.65	-1.45	-0.26	-5.39***
	(0.37)	(0.03)	(0.21)	(0.14)	(0.05)	(0.10)	(0.18)	(0.59)	(0.01)
	[-44.3%]	[-75.7%]	[100.0%]	[69.6%]	[-46.2%]	[-70.3%]	[-46.1%]	[-3.3%]	[-69.4%]
April	-0.14	1.45***	-0.14	0.67	-0.43	-0.24	0.56	0.58	-0.73**
	(0.41)	(0.01)	(0.82)	(0.14)	(0.90)	(0.28)	(0.26)	(0.12)	(0.05)
	[-8.7%]	[80.0%]	[-19.5%]	[592.0%]	[-11.8%]	[-4.8%]	[110.0%]	[51.1%]	[4.0%]
May	0.26**	0.86***	0.01	0.17	0.20**	0.25*	0.14	0.56***	0.14
	(0.03)	(0.01)	(0.36)	(0.15)	(0.02)	(0.07)	(0.14)	(0.01)	(0.12)
	[928.0%]	[481.0%]	[48.9%]	[334.0%]	[149.0%]	[182.0%]	[218.0%]	[895.0%]	[129.0%]

This table estimates the change in number of fires detected per 100 km^2 in each administrative unit from 2017 to 2020. The value in each cell is the treatment effect, with the p-value in brackets and the percentage change in squared brackets.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Effect on fire outcomes during strict enforcement (2017–2020): detailed satellite breakdown

Period	Total fires	Aqua			Terra		
		Total	Night	Day	Total	Night	Day
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Jan–May	-3.24^{*}	-3.32^{**}	-0.05^{**}	-3.50^{**}	-0.16	0.06	-0.26
	(0.08)	(0.04)	(0.03)	(0.02)	(0.18)	(0.13)	(0.13)
January	-0.07	-0.02	0.00	-0.03	-0.03	-0.00	-0.02
	(0.45)	(0.49)	(0.28)	(0.52)	(0.34)	(0.20)	(0.36)
February	-0.62	-0.53	-0.03	-0.49	-0.11	-0.06	-0.05
	(0.19)	(0.22)	(0.17)	(0.22)	(0.19)	(0.25)	(0.23)
March	-2.42^{*}	-2.14^{*}	0.02	-2.18^{*}	-0.29	0.03	-0.21
	(0.08)	(0.07)	(0.10)	(0.06)	(0.16)	(0.16)	(0.11)
April	0.18	-0.22	0.07^{*}	-0.30	0.36^{*}	0.24^{*}	0.03
	(0.14)	(0.14)	(0.08)	(0.15)	(0.08)	(0.05)	(0.20)
May	0.29^{**}	0.19^{**}	0.00	0.19^{**}	0.07^{**}	0.02	0.05^{**}
	(0.01)	(0.01)	(0.18)	(0.01)	(0.01)	(0.16)	(0.04)

This table estimates the change in number of fires detected per 100 km^2 in an administrative unit from 2017 to 2020. Aqua and Terra refer to the MODIS satellites, with separate breakdowns for nighttime and daytime detections. MODIS fires above the 30% confidence threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Donor weights for synthetic control by province - main specification

Donor unit	**Country**	**Chiang Rai**	**Chiang Mai**	**Mae Hong Son**	**Tak**	**Phayao**	**Nan**	**Lampang**	**Phrae**	**Lamphun**
Bawlake	MN	-	-	0.21	-	-	-	-	-	-
Danu SAZ	MN	-	0.40	-	-	-	-	-	-	-
Det Khi Na	LA	-	-	-	-	-	0.14	-	-	0.13
Hpa-An	MM	0.59	-	-	-	-	-	-	-	-
Hpapun	MM	-	0.12	-	0.28	-	-	-	0.26	0.11
Langkho	MM	-	-	0.20	-	0.15	-	-	-	0.24
Louangnamtha	LA	-	0.16	0.26	-	-	-	-	-	-
Myawaddy	MM	-	-	-	0.26	-	-	-	-	-
Narathiwat	TH	-	-	-	-	0.21	-	-	-	-
Pyinoolwin	MM	0.32	-	-	-	-	-	-	-	-
Ranong	TH	-	0.15	-	-	-	-	-	-	-
Rayong	TH	-	-	-	-	-	-	-	-	0.13
Taunggyi	MM	-	-	-	0.12	-	-	-	-	-
Taungoo	MM	-	-	-	-	-	-	-	0.15	-
Uttaradit	TH	-	-	-	-	-	-	1.00	0.41	0.39
Xaignabouly	LA	-	-	0.14	-	0.18	0.71	-	-	-
Yala	TH	-	-	-	0.13	-	-	-	-	-
Yamethin	MM	-	-	-	-	0.37	-	-	-	-

Notes: Country codes: MM (Myanmar), TH (Thailand), LA (Laos). A dash (-) indicates donor weights < 0.10. SAZ refers to Self-Administrative Zone. This table refers to weights used for the estimates from January to May presented in Table 2.

Table 13: Effect on *total fires detected by VIIRS*

Period	Strict (2017-2020)	Weak (2013-16)
Jan-May	−14.565	−5.921
	(0.175)	(0.187)
January	0.635	0.400
	(0.235)	(0.349)
February	−4.678	−2.024
	(0.199)	(0.205)
March	−2.172	5.424
	(0.337)	(0.380)
April	2.402	6.679
	(0.404)	(0.386)
May	2.110	0.968
	(0.193)	(0.253)

This table estimates the change in number of total fires as detected by the VIIRS sensor, per 100km^2 in an administrative unit following the ban. VIIRS fires above the 30 percent threshold are used. The weights are determined by MODIS fires, shown in Table 12.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Effects depending on crop production during strict enforcement (2017–2020)

Period	Q1	Q2	Q3	Q4
Jan-May	−1.96*	−1.49	−0.39	−0.11
	(0.07)	(0.11)	(0.21)	(0.11)
January	0.04	−0.04	−0.05	−0.03
	(0.25)	(0.42)	(0.24)	(0.58)
February	−0.07	−0.15	−0.12	−0.08
	(0.30)	(0.36)	(0.13)	(0.19)
March	−1.50*	−0.96	−0.08	−0.04
	(0.08)	(0.14)	(0.24)	(0.23)
April	−0.14	0.02*	0.10*	0.01
	(0.24)	(0.09)	(0.06)	(0.24)
May	0.08*	0.11**	0.06*	0.03**
	(0.07)	(0.02)	(0.07)	(0.03)

This table estimates the change in fires per 100km^2 in an administrative unit following the ban during strict enforcement (2017–2020). The columns refer to the quantile of crop production, with Q1 being the lowest (lowest crop production in the country), based on SPAM (International Food Policy Research Institute, 2019). MODIS fires above the 30 percent threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15: Effects depending on dominant crop during strict enforcement (2017–2020)

Period	Maize	Rice	Cassava
Jan-May	−0.029**	−1.375**	−0.131
	(0.018)	(0.048)	(0.133)
January	−0.003*	−0.090	0.004
	(0.066)	(0.169)	(0.175)
February	−0.005*	−0.425	−0.011
	(0.072)	(0.139)	(0.331)
March	−0.013**	−0.667*	−0.140**
	(0.048)	(0.096)	(0.018)
April	−0.002	0.057	−0.006
	(0.807)	(0.151)	(0.193)
May	0.000*	0.105**	−0.007
	(0.054)	(0.030)	(0.133)

This table estimates the change in fire outcomes per 100km^2 in an administrative unit following the ban during strict enforcement (2017–2020). The columns refer to the effect on fires taking place in regions depending on the type of crop with the highest level of production in the pixel, based on (International Food Policy Research Institute, 2019). MODIS fires above the 30 percent threshold are used.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: Effects of fire intensity during strict enforcement (2017–2020)

Period	Low intensity	High intensity
Jan-May	−1.20*	−2.51*
	(0.05)	(0.07)
January	−0.05	−0.03
	(0.43)	(0.49)
February	−0.34	−0.27
	(0.11)	(0.27)
March	−0.65*	−1.90*
	(0.08)	(0.08)
April	0.24	−0.19
	(0.10)	(0.13)
May	0.18**	0.10***
	(0.02)	(0.01)

This table estimates the change in fires per 100km^2 in an administrative unit following the ban during strict enforcement (2017–2020). Low intensity fires refer to fires with below median fire radiative power, and high intensity fires refers to fires with above median fire radiative power across the entire sample.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Robustness tests: effect on total fires during strict enforcement period (2017–2020)

2*Period	Robustness Tests								
	Baseline	Pre-2008-2012	No Repeated	High Confidence	No Myanmar	No Laos	No Thailand	No Climate	Fixed Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Jan–May	-3.24^{*}	-3.51^{*}	-3.10^{**}	-1.85^{*}	-3.17^{*}	-3.21^{**}	-3.37^{*}	-3.00^{**}	-3.24^{**}
	(0.08)	(0.07)	(0.04)	(0.07)	(0.08)	(0.03)	(0.06)	(0.04)	(0.04)
January	−0.07	−0.02	−0.07	−0.04	−0.09	0.00	−0.11	−0.05	0.02
	(0.45)	(0.39)	(0.45)	(0.33)	(0.29)	(0.44)	(0.43)	(0.52)	(0.59)
February	−0.62	−0.53	−0.56	−0.23	−0.27	−0.59	−0.55	−1.08	−0.19
	(0.19)	(0.26)	(0.14)	(0.33)	(0.70)	(0.12)	(0.15)	(0.18)	(0.77)
March	-2.42^{*}	-2.39^{*}	-2.05^{*}	-1.46^{*}	−1.49	-2.75^{**}	-2.89^{*}	-2.23^{*}	−1.74
	(0.08)	(0.06)	(0.06)	(0.07)	(0.16)	(0.03)	(0.07)	(0.07)	(0.25)
April	0.18	−0.03	0.09	−0.10	0.05	0.08	−0.28	0.08	−1.50
	(0.14)	(0.16)	(0.10)	(0.13)	(0.15)	(0.12)	(0.19)	(0.16)	(0.37)
May	0.29^{**}	0.26^{**}	0.24^{***}	0.09^{**}	0.19^{**}	0.28^{**}	0.27^{**}	0.30^{**}	0.17^{*}
	(0.01)	(0.01)	(0.01)	(0.03)	(0.01)	(0.01)	(0.02)	(0.01)	(0.05)

This table shows the treatment effects during the strict enforcement period (2017-2020) across different robustness tests. (1) Baseline specification; (2) Pre-intervention period 2008-2012 instead of 2004-2012; (3) Excluding repeated fires, by excluding fires taking place in the same 1 km^2 grid cell in the same day or in consecutive days or in the same day; (4) Only including fires with above 75% confidence, instead of the baseline 30% threshold; (5) Excluding donor units from Myanmar; (6) Excluding donor units from Laos; (7) Excluding donor units from Thailand; (8) Excluding climatic variables from the synthetic control weights generation; (9) Fixing the set of weights to the one generated for the January to May estimation.
p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

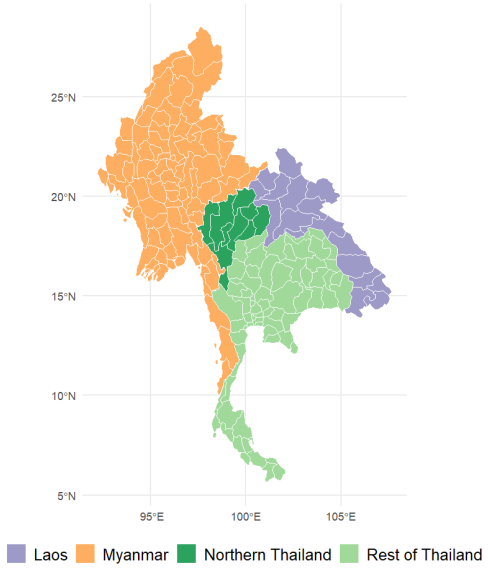


Figure 6: Administrative units for the study area, by country, and within Thailand, by splitting into provinces in the Northern region that had a seasonal burning ban and the rest of the country.

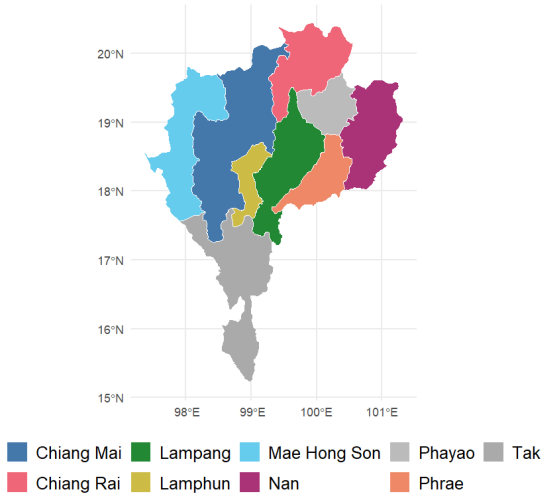


Figure 7: The nine treated provinces in Northern Thailand.

Table 18: Backdating tests: estimated treatment effects on total fires

2Period	Backdate to 2009	Backdate to 2010	Backdate to 2011
	(1)	(2)	(3)
Validation Period (Backdating year - 2012)			
Jan–May	−0.55	0.46	1.58
	(0.46)	(0.55)	(0.36)
January	−0.14	−0.01	0.02
	(0.21)	(0.66)	(0.47)
February	0.32	0.98	1.75
	(0.23)	(0.14)	(0.08)
March	−0.63	−0.24	−0.12
	(0.80)	(0.62)	(0.47)
April	−0.17	−0.24	−0.46
	(0.40)	(0.40)	(0.19)
May	−0.03	−0.05	−0.04
	(0.30)	(0.28)	(0.43)
Weak Enforcement (2013–2016)			
Jan–May	−1.34	−1.03	−0.99
	(0.28)	(0.29)	(0.27)
January	−0.09	−0.07	−0.07
	(0.41)	(0.53)	(0.52)
February	−0.12	−0.06	−0.07
	(0.45)	(0.45)	(0.43)
March	−0.76	−0.65	−0.61
	(0.45)	(0.34)	(0.33)
April	0.27	0.25	0.23
	(0.35)	(0.37)	(0.30)
May	0.03***	0.03**	0.04***
	(0.19)	(0.22)	(0.18)
Strong Enforcement (2017–2020)			
Jan–May	−3.36*	−3.10*	−3.07*
	(0.10)	(0.08)	(0.07)
January	−0.17	−0.13	−0.13
	(0.30)	(0.40)	(0.41)
February	−0.61	−0.53	−0.53
	(0.24)	(0.21)	(0.21)
March	−2.64	−2.69	−2.64*
	(0.16)	(0.11)	(0.08)
April	−0.14	−0.04	−0.06
	(0.15)	(0.18)	(0.16)
May	0.23**	0.21***	0.23***
	(0.02)	(0.01)	(0.01)

This table shows the estimated treatment effects and corresponding p-values from the backdating tests. The synthetic control is constructed using pre-treatment data up to the year specified in the column heading. The "Validation Period" section shows the estimated effect of the "treatment" for the years between the backdate and the actual treatment (2013), before the beginning of the ban.

P-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

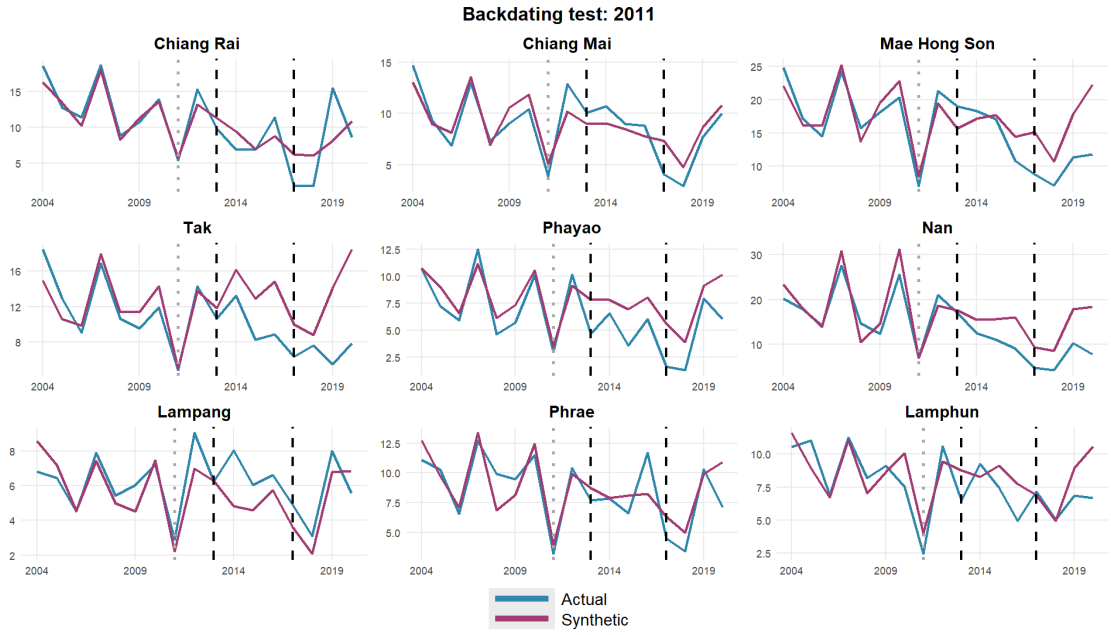


Figure 8: Backdating tests - 2011. This figure shows the actual values for each province in Northern Thailand, compared to the synthetic one, with the synthetic controls being generated based on 2004-2011 data rather than 2004-2012.

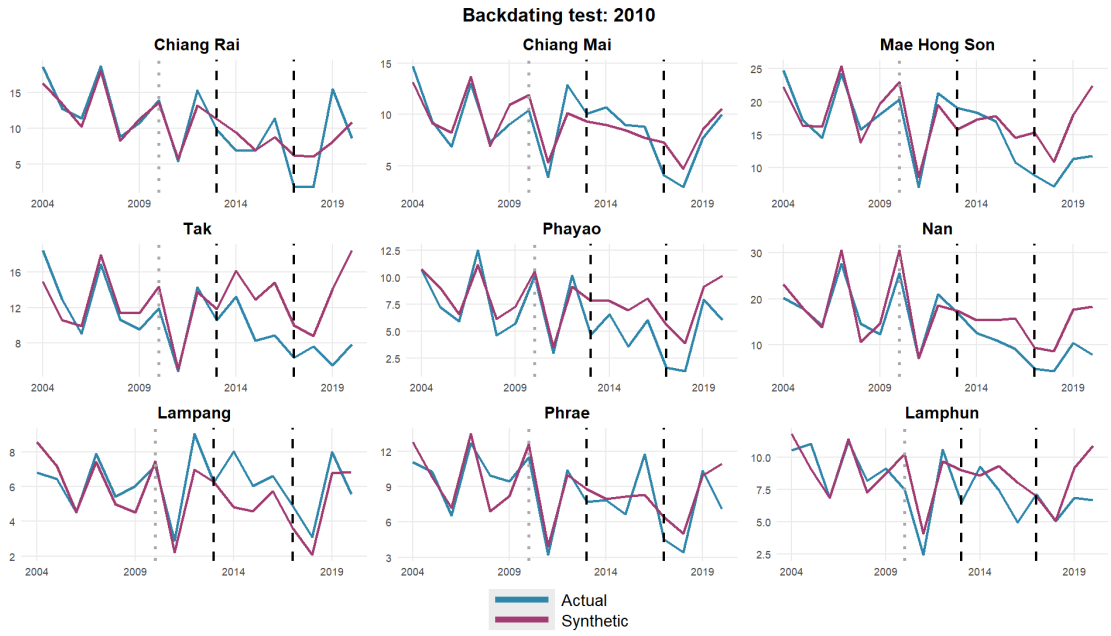


Figure 9: Backdating tests - 2010. This figure shows the actual values for each province in Northern Thailand, compared to the synthetic one, with the synthetic controls being generated based on 2004-2010 data rather than 2004-2012.

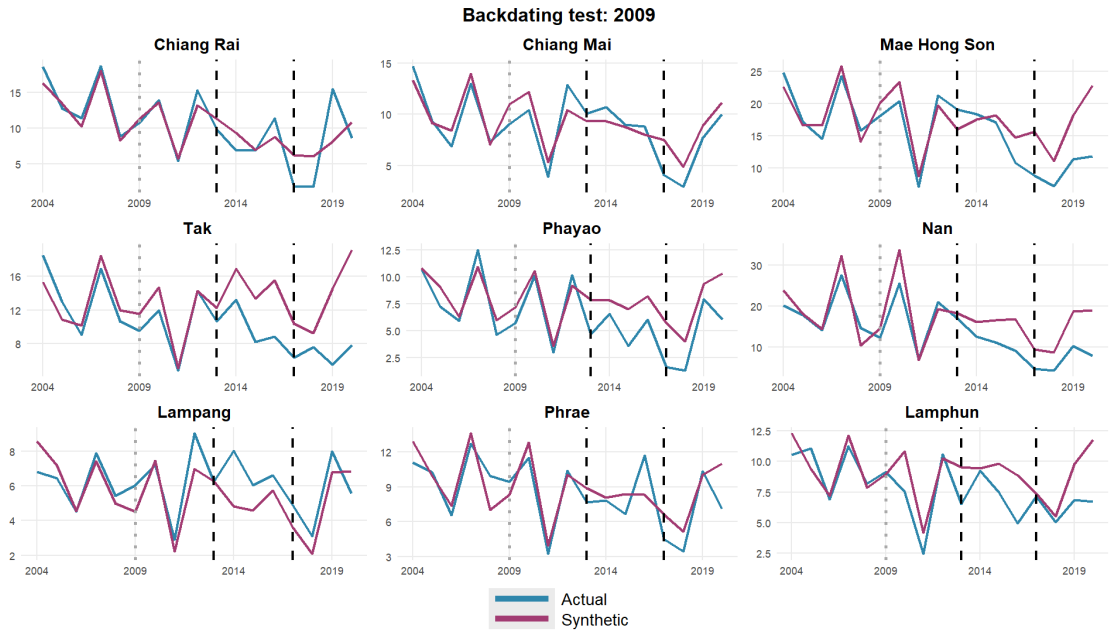
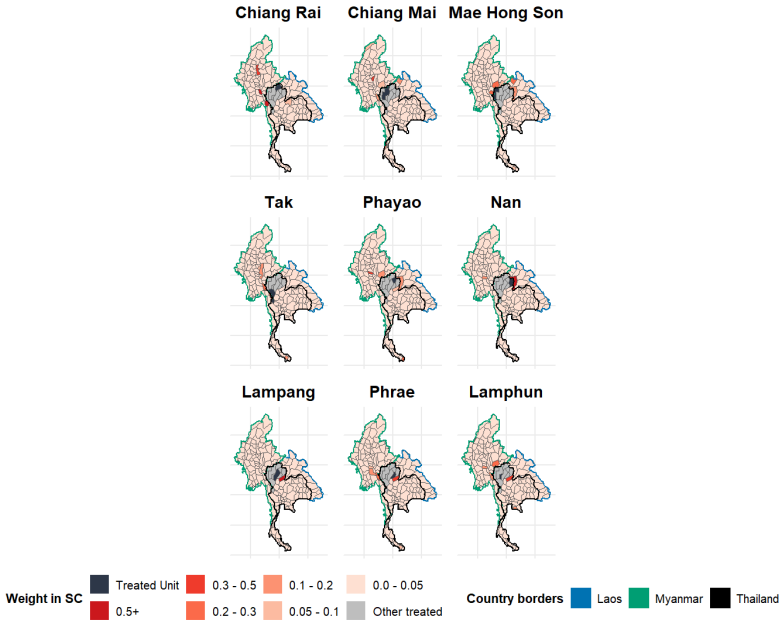


Figure 10: Backdating tests - 2009. This figure shows the actual values for each province in Northern Thailand, compared to the synthetic one, with the synthetic controls being generated based on 2004-2009 data rather than 2004-2012.

Figure 11: Synthetic control weights. Corresponds to Figure 3, with the full map.



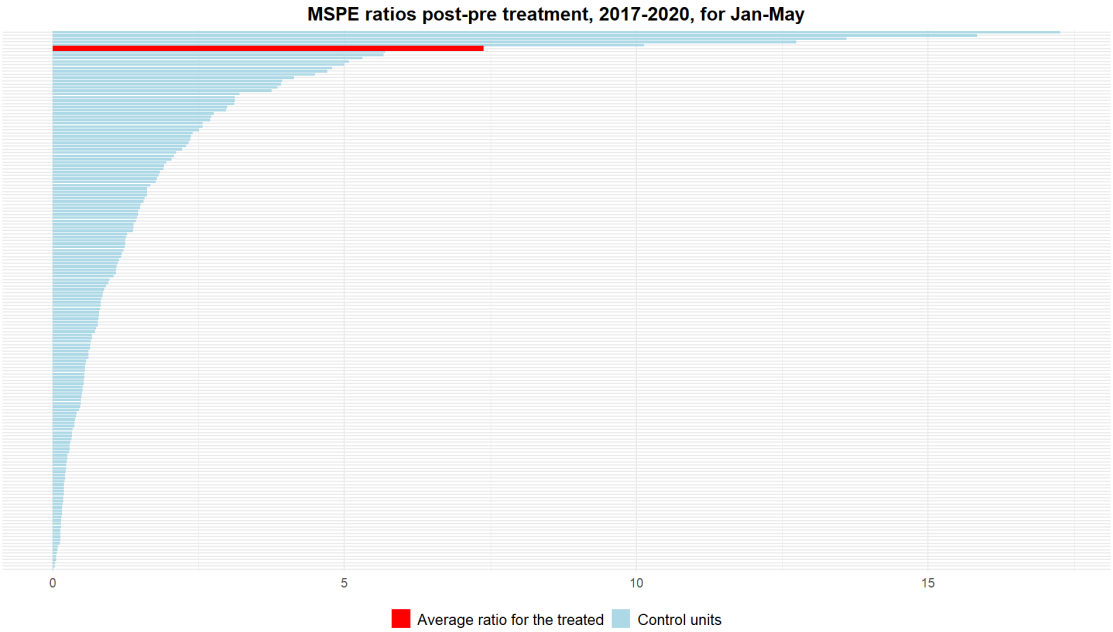


Figure 12: Ratio of mean squared prediction errors (MSPE) of the main specification, examining the ratio of MSPEs in the period with strict enforcement (2017-2020) to the MSPE prior to the treatment (2004-12). The average MSPE of the treated units is compared to all control units. This comparison focuses on the entire period of January to May. It is used to compute the p-value.

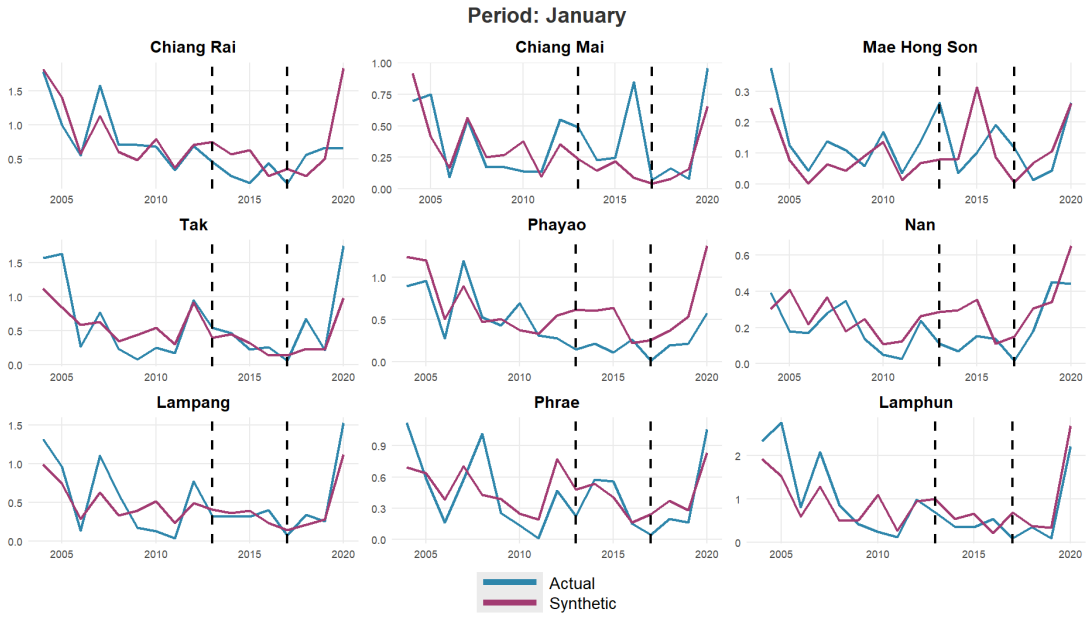


Figure 13: Synthetic control and actual fire values in January every year by province.

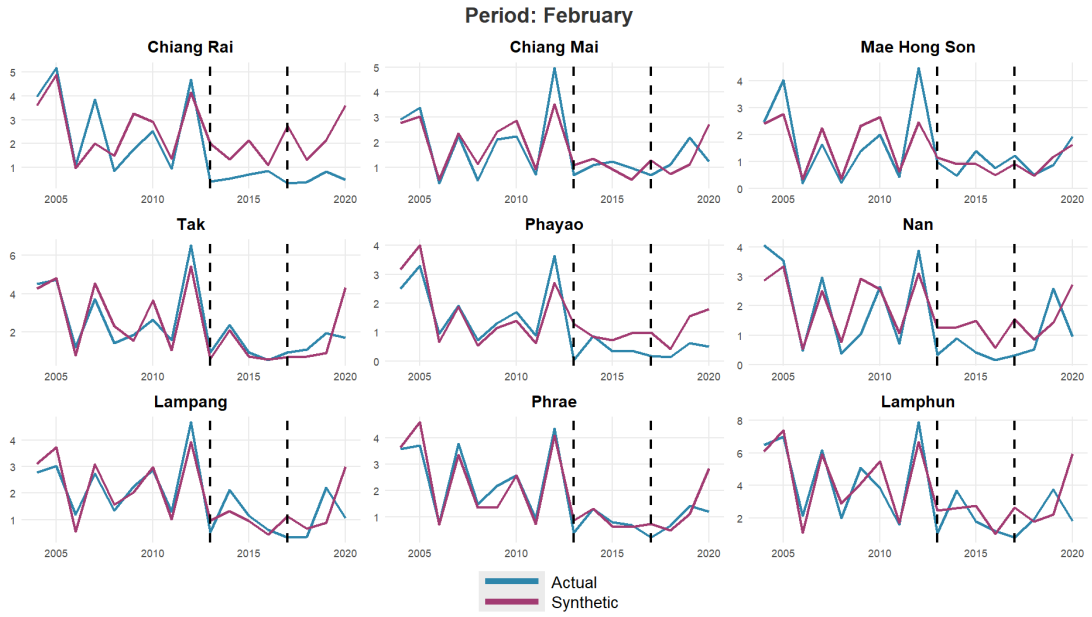


Figure 14: Synthetic control and actual fire values in February every year by province.

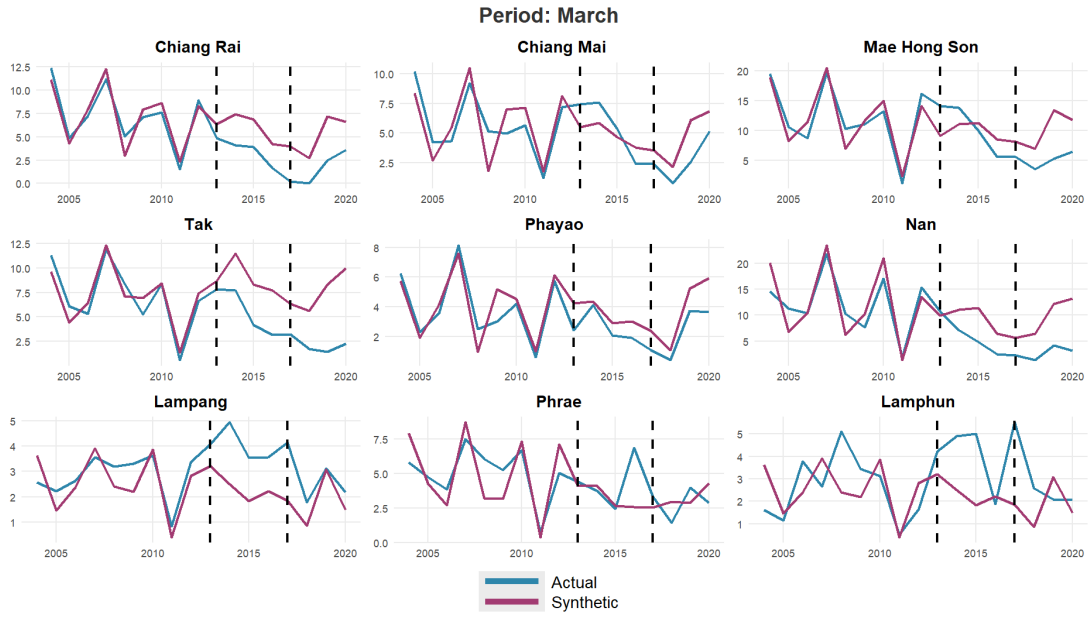


Figure 15: Synthetic control and actual fire values in March every year by province.

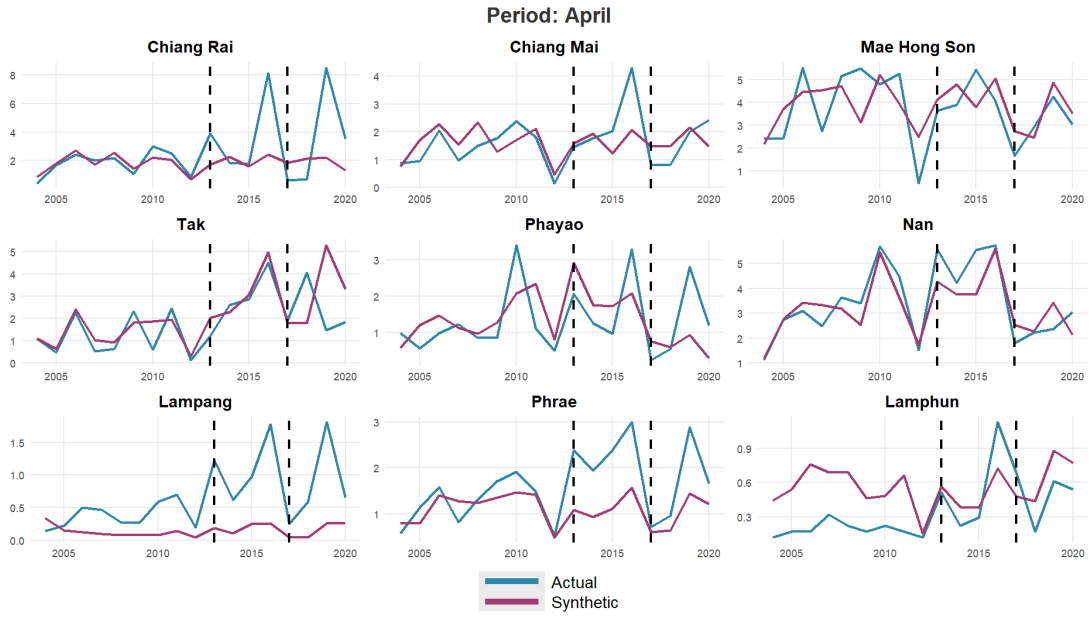


Figure 16: Synthetic control and actual fire values in April every year by province.

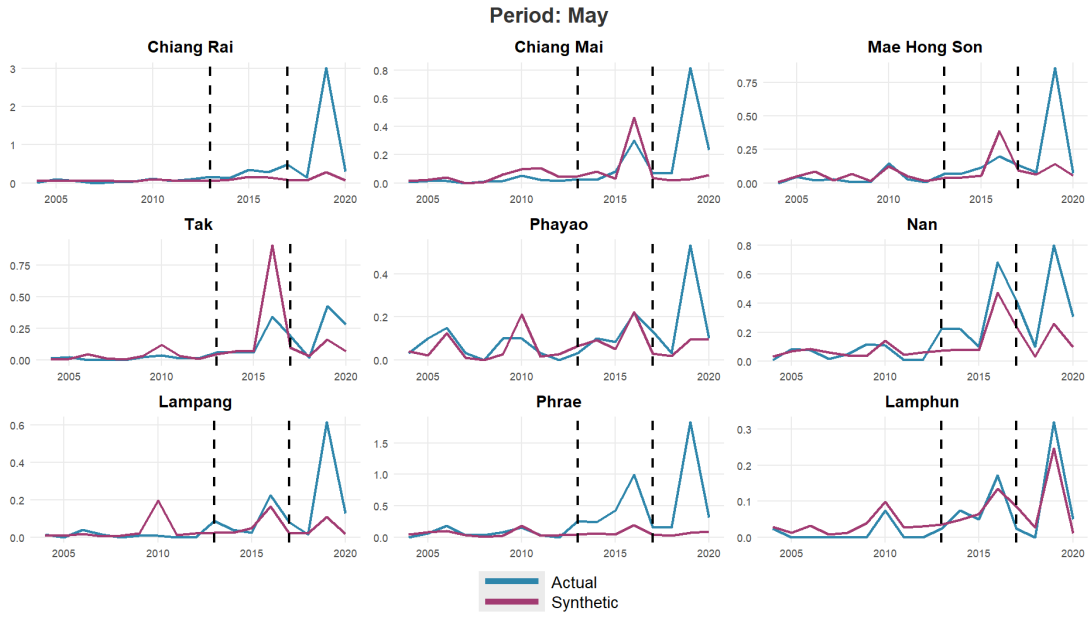


Figure 17: Synthetic control and actual fire values in May every year by province.